

Master Beekeeping Part I: Honey Bee Evolution, Biology, and Behavior

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Homepage

Video Transcript

Honey bee biology and behavior. Why is this even important for me to know as a beekeeper? You might be itching for the hands on practical learning aspects of the course, but it's really important to understand these organisms first and foremost. Because a master beekeeper is a combination of the practical aspects of managing a colony, combined with a deep understanding of honey bees as living organisms. And before you can start to perform tasks like making splits or moving your colonies or identifying and treating different diseases, you have to know important things such as the developmental time for a worker honey bee.

I'm really excited to have you learn the inner workings of a bee society and why honey bees behave the way they do. In learning about bees truthfully just inspires us to become better beekeepers. I know this because it happened to me. I started to see honey bees as more than just crop pollinators for human food and for honey producers. And learning about bees just really helps you garner a deeper respect for honey bees as complex organisms.

What you'll do

- **Explore the evolution of the honey bee with a focus on how its social behavior evolved**
- **Understand how honey bees reproduce**
- **Identify critical processes in honey bee development**
- **Understand how honey bees communicate**
- **Explore honey bee biology and physiology**
- **Understand how honey bees construct and maintain their hives**



Course Description

In order for your colonies to truly thrive, you must have a deep understanding of honey bees as living organisms. Knowing how honey bees have evolved and understanding why they behave the way they do



is key in supporting your colonies' survival and success. Discover the “what” and “why” of honey bee behavior from experts at Cornell University so you can better identify and respond to the needs of your honey bees.

Course Authors



Emma Walters, MSc
Senior Honey Bee Extension Associate, Cornell University

Emma Kate Walters works with beekeepers to understand factors affecting honey bee health, with a focus on parasites, diseases, pesticides, and management practices. She is particularly interested in training beekeepers in methods that improve their colonies' health. At Cornell University, she develops and oversees beekeeping programs such as the NYS Beekeeper Tech Team, the Cornell University Master Beekeeping Program, and the Beekeeping Essentials course. Emma started beekeeping in 2011 in Ontario, Canada, and completed her Master's degree investigating honey bee behavioral genetics at Western University in Canada.



Eleanor Andrews, MSc
Honey Bee Extension Assistant, Dyce Lab of Honey Bee Studies, Cornell University

Eleanor Andrews is a Honey Bee Extension Assistant at Dyce Lab for Honey Bee Studies and a Ph.D. candidate in Development/Rural Sociology at Cornell University. Ellie is interested in strengthening beekeepers' ability to evaluate sources of information about beekeeping, which is especially important now that honey bee health has moved into the political arena. Her research addresses how beekeepers learn to keep bees, how beekeeping clubs support different forms of sustainable beekeeping, and how educational resources for beekeepers have developed over time. She believes that the problems that beekeepers face can only be addressed by combining physical science with social science. Ellie also brings a focus on pedagogy and organizational frameworks to support learning, based on her experiences teaching at Cornell, Peace Corps, a Quaker boarding school, and prison.

Thomas D. Seeley, Ph.D.
Horace White Professor in Biology, Department of Neurobiology and Behavior, Cornell University





Thomas D. Seeley, biologist and writer, is a professor in the Department of Neurobiology and Behavior at Cornell University. He teaches courses on animal behavior and does research on the behavior, social life, and ecology of honey bees. Professor Seeley is an avid beekeeper and began keeping bees as a high school student, when he shook a swarm into a box and brought it home. His scientific work is summarized in four books: "Honeybee Ecology" (1985), "The Wisdom of the Hive" (1995), "Honeybee Democracy" (2010), and "Following the Wild Bees" (2016). In

recognition of his scientific contributions, he has been honored by an Alexander von Humboldt Distinguished U.S. Scientist Award, awarded a Guggenheim Fellowship, and elected a fellow of the American Academy of Arts and Sciences. Professor Seeley writes: "These awards are gratifying, but for me the most important prizes by far are the discoveries that I have made about the inner workings of honey bee colonies."



Christina Wahl, Ph.D.
Courtesy Assistant Professor, Department of Biomedical Sciences, Cornell University

A fourth-generation beekeeper from upstate New York who grew up working in her family's commercial beekeeping business, the Thousand Islands Apiaries, **Christina Wahl** is a physiologist and developmental

biologist with technical expertise in microscopic surgery, microscopic anatomy, and histology. She has 30 years of research experience, much of it supported by the National Institutes of Health. Professor Wahl has taught at Cornell University in Ithaca, NY, and at Wells College in Aurora, NY. She has been a volunteer instructor for the NY Bee Wellness program and has both taught and mentored beginning beekeepers in the Finger Lakes region. Professor Wahl has designed and co-taught workshops on queen rearing. She directs the educational exhibit on honey bees and pollinators at the New York State Fair on behalf of the Empire State Honey Producers. Professor Wahl is the editor of the book "Morphometrics," published by InTech. She has authored several book chapters and many original peer-reviewed research papers and is a member of several professional science organizations. Currently, Professor Wahl is engaged in a study entitled "Number and distribution of multiple foundress mites *Varroa destructor* in northern New York drone brood combs during spring and fall." She is actively engaged in efforts to understand and prevent health problems of the honey bee.



Discussion: Introduce Yourself

One of the benefits of this course is making connections and learning from the experiences of fellow beekeepers. Using the discussion below, please tell us about yourself!

- Where are you located?
- How did you get interested in beekeeping?
- How many years of experience do you have?
- What are your beekeeping goals?

Instructions:

- Click **Reply** to post a comment or reply to another comment.
- Please respond in a text format or as a video using the filmstrip icon that is available once you click **Reply**. (If posting a video response, we recommend that you do not use your cell phone as most do not use Flash software, which is required to convert the recording.)

Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [Cornell's policy regarding plagiarism](#) (the presentation of someone else's work as your own without source credit).

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Read: Preview Your Course Project

At the end of this course, you will practice applying the teachings in a course project. You can review all relevant information on the **Syllabus** page of this course.

☆ Key Points

You will apply the content from this course in a final project.

Since each module builds on the previous ones, it is essential that you work through the course in the order it appears.

Note: Though your work will only be seen by eCornell, you should take care to obscure any information you feel might be of a sensitive or confidential nature.

Course Project Part

Preview the course project below. As you work through the course, reflect on how these topics relate to the course content and to your experience.

— Course Project — Creating an Outline for a Presentation

In this project, you will create an outline for a hypothetical 45-minute presentation. The presentation can be on any subject introduced in this course. You will need to conduct research using books, reputable websites from scientific organizations or extension personnel, and/or beekeeping magazines as resources to help you find information to include.

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Module Introduction: **The Evolution of the Honey Bee as a Social Insect**



Honey bees are one of the most fascinating animals on Earth. While an individual honey bee cannot survive on its own, a colony of honey bees, working together as a cooperative unit, can thrive. In this module, you will explore how honey bees evolved from solitary wasps into one of the most advanced examples of social living on the planet.

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Watch: The Honey Bee Colony as a Superorganism

A honey bee colony is a captivating thing. A mature colony in the peak of summer can contain 30,000 to 50,000 individuals, all working together as a cooperative unit that functions as an integrated whole.

The individual members cannot survive on their own. The queen, who is the mother of all the bees in a colony, depends on her worker daughters for building and defending the nest, keeping it stocked with food, rearing the brood, fighting parasites and pathogens, and everything else. Meanwhile, the workers depend on their queen mother to lay all the eggs the colony needs to produce its workers, drones, and young queens.

As a superorganism, the bees are able to benefit from social living. They forage very efficiently by communicating the location of rich food sources to their nest mates, they regulate the temperature of their home, and they reproduce through swarming. Swarming is a remarkable example of a colony's social integration, because it requires finding a new home site that all of the bees agree upon. For successful beekeeping, it is important to understand these and other traits that have evolved with the social life of these bees.

Watch the video to learn about some more of the social behaviors honey bees have evolved that make them such successful organisms.

Video Transcript

So thousands of individual organisms are working together and acting like one large organism, or a superorganism, and they're cooperatively collecting food, they're mounting defenses against intruders and they're coordinating behavioral responses to a changing environment. So in order to do all of this socially, complex communication is required. And honey bees use three forms of communication: acoustic — touch and vibration, chemical and visual. And the most famous form of communication with honey bees — the waggle dance, actually combines some of these three forms.

Another really neat thing about this social behavior in honey bees is social immunity. So diseases can be easily transmitted when we have lots of individuals in close proximity to

☆ Key Points

A superorganism is a group of related, social individuals working together to effectively cooperate as one large organism.



each other, like honey bees in a hive. And so, social immunity is using social behaviors to try to reduce the disease spread or reduce parasites that are in the colony. So honey bees some of them may groom each other to remove mites from each other's backs, or they can detect brood that's infected with diseases and actually remove them from the colony, a form of hygienic behavior. And for some infections like chalkbrood infections, they can actually develop a social fever. And this is where they physically raise the temperature of the brood to the point where it gets too hot for the chalkbrood infection to live. So these are just some of the really cool things that these very complex socially behaving honey bees are doing.

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Read: Other Superorganisms

Honey bees are not alone in this style of advanced social living in which offspring "stay at home" for life to live with one or both of their parents.

Termites, ants, some wasps, naked mole rats, and a few other organisms also live this way. These social colonies are often referred to as superorganisms because they are many individual organisms that function together as one large organism. Sometimes the behavior of an individual member of the colony may seem confusing or counterintuitive, but makes sense when it is understood as part of the functioning of a superorganism.

Photo: Giant nest built by termites (an example of a superorganism). Photo credit: Graham Thompson



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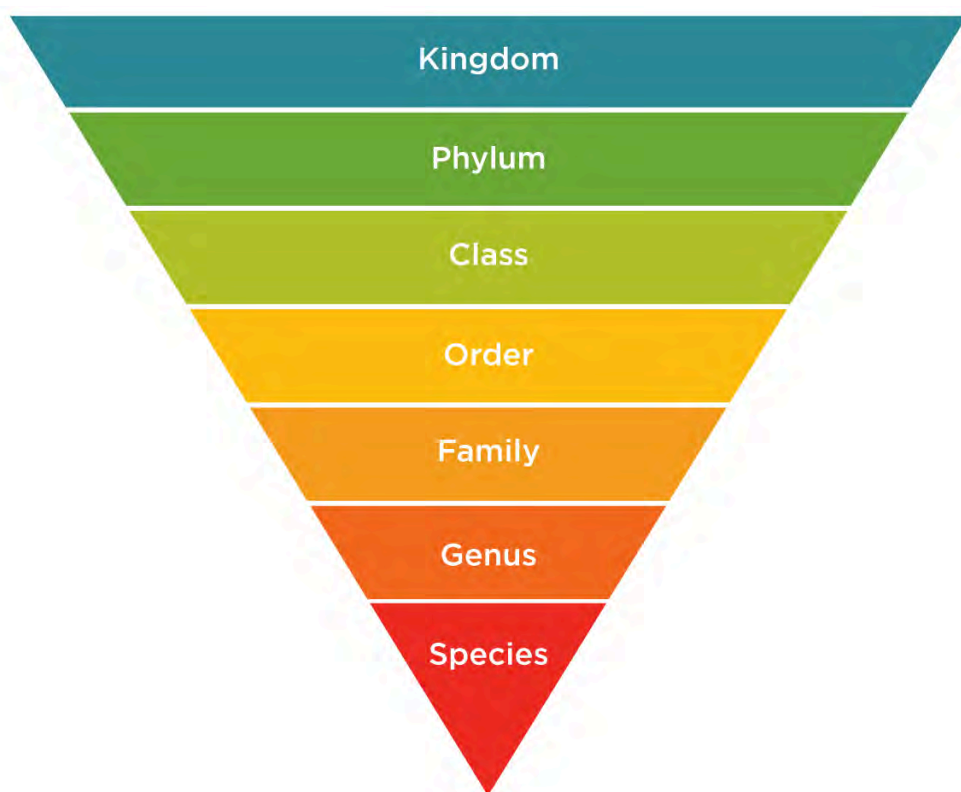
Read: What Does It Mean to Be a Honey Bee?

All living organisms can be classified scientifically into different categories to help us understand which characteristics they share and which are different. We give organisms scientific names based on their place in this classification system. This system is useful in understanding why Western honey bees are called *Apis mellifera*, identifying how they are unique compared to other bees and insects, and describing the immense diversity of living organisms we see around us.

☆ Key Points

Western honey bees are referred to as *Apis mellifera* because of how they are classified

As a Master Beekeeper, you need to be able to distinguish between honey bees and the other insects they are frequently confused with



This classification system is broken down into the following groups: kingdom, phylum, class, order, family, genus, and species. As we move down the groups from kingdom to



species, we observe more similarities among organisms' appearances, behaviors, and genetics.

How does the honey bee fit in with the rest of life on Earth?

Kingdom

At its most basic level, life is classified into five different kingdoms: animals, plants, fungi, bacteria, and archaea (which are other small, single-celled organisms). Honey bees, along with all other insects, are in the animal kingdom.

The five kingdoms of life:



Animals



Plants





Fungi



Bacteria



Archaea

Phylum



The next division is at the phylum level. There are many phyla (plural of "phylum") within the animal kingdom. Honey bees are members of the phylum called Arthropoda, so we call them arthropods. All arthropods are invertebrates, as they do not develop a backbone and instead have a tough outer body shell called an exoskeleton. Arthropods have segmented bodies and pairs of jointed appendages.

Some members of the phylum Arthropoda:





Class

The next division is at the class level. Within Arthropoda, there are several classes, including Insecta, Arachnida (eg., spiders and *Varroa* mites), and Crustacea (e.g., crustaceans), to name a few. Honey bees belong to the class Insecta (Insects), which is the most diverse class of animals on Earth. Insects are defined by having six legs across three body parts (head, thorax, and abdomen), compound eyes, and a pair of antennae.

Some members of the class Insecta:







Order

Within the class Insecta, there are many different orders of insects. Bees are in the order Hymenoptera, which includes ants, bees, and wasps. Insects in the order Hymenoptera usually have two pairs of wings joined together by small hooks. Females have a hardened ovipositor (egg-laying organ) that may be used for piercing, stinging, or reproduction. They are also the biggest-brained insects, hence they are considered the "intelligentsia" of the insect world.

Aside from similarities in body structure, some Hymenopterans share other traits with honey bees: they have similar social living strategies with a queen and cooperative helpers, they build nests and store food, and they sting to defend their colonies. Some characteristics can cause people to confuse other bees and wasps with honey bees.

Some members of the order Hymenoptera:







Family

Within the order Hymenoptera, there are nine families of bees that consist of about 20,000 documented bee species throughout the world! Honey bees are within the family Apidae. All bees within this family carry pollen on their legs. It is the largest family of bees and includes bumble bees, carpenter bees, and many others in addition to honey bees.

In addition to honey bees, large carpenter bees (left) and bumble bees (right) also belong to the family Apidae.







Photo credit: Bee Informed Partnership

Genus

The last two categories comprise the species name. Every organism has a two-part species name. The first part is the genus name and the second part is the species name. Bumble bees, for instance, are all part of the genus '*Bombus*', while honey bees are all part of the '*Apis*' genus. Organisms within the same genus often look similar to each other.

Species



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The Western honey bee has *Apis mellifera* for its species name. What does "mellifera" mean? Melli means "honey" and fera means "bearer", so it was named as the "bearer" or maker of honey.

There are eight species of honey bee (members of the genus *Apis*) found throughout the world, and some that you may be familiar with are the Eastern honey bee (*Apis cerana*), the dwarf honey bee (*Apis florea*), and the giant honey bee (*Apis dorsata*). These different species have different body sizes, behaviors, and nest designs, but they all have a 'honey bee' look to them with similar colors and banding patterns, and they all make honey. Most importantly, organisms need to be the same species in order to reproduce. Western honey bees can only mate and create offspring with other Western honey bees. However, closely related species can share parasites and diseases. *Apis cerana* is the original host of *Varroa* mites and *Nosema ceranae*, but these parasites are able to thrive in colonies of the closely related Western honey bee.

Different species of honey bees (from left to right): *Apis dorsata*, *Apis florea*, and *Apis mellifera*.





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Read: Distinguishing Honey Bees From Other Insects

Some insects are often confused with honey bees, especially wasps, other bees, and flies. As a Master Beekeeper, you must be able to distinguish honey bees from other organisms. You may receive calls from the public about swarms and unwanted nests, and it is important to identify whether the insects in question are actually honey bees in order to determine how you can be of service. If you receive one of these calls, ask the person to describe the nest and the appearance of the insect, and to send a photo if they have one. If the insect in question is a wasp and a nuisance, you can refer them to a pest control company. However, if the location of the colony is not a threat to anyone, you can suggest they let the colony be, as wasps are important natural enemies of many insect pests.



Distinguishing Honey Bees From Other Insects

Organism	Nest	Appearance	Food
Yellow Jackets (Wasps) 	Some species nest underground; others build round nests covered in papier-mâché-like material in high locations; no honey is produced or stored in the nest	Bright yellow and black banding; hairless, slender waist; often have yellow legs; social colonies of many individuals; body is about the same size as honey bee	Adults feed on nectar; larvae are carnivores
Paper Wasps (Wasps) 	Open 'papery' hexagonal cells; no honey is produced or stored in the nest	Yellow and black or brown and black banding; hairless, slender waist; social colonies of many individuals; body is often larger than honey bees	Adults feed on nectar; larvae are carnivores



Syrphid Flies



Do not create nests; do not live socially

Yellow and black or brown and black banding; single pair of wings that don't fold when resting; large eyes that almost touch; some hair on head and thorax; very short antennae; body can be smaller than or similar in size to honey bees

Adults feed on pollen and nectar; larvae are carnivores

Paper wasp photo credit: Sarah Bluher

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Quiz: Distinguishing Honey Bees From Other Insects

In the following exercise you will have to decide whether a photograph is a honey bee or another insect.

You've received several photos from the public inquiring about bees on their properties. You quickly realize however, that some of these photos are not bees at all. Identify which photos are honey bees, syrphid flies, and wasps.

You must achieve a score of 100% on this quiz to complete the course. You may take the quiz as many times as you need to achieve that score.

Please complete this quiz in the course.

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Read: The Evolution of Honey Bees

How did bees evolve? Before bees existed, there were several species of wasps. These wasps, just like wasps today, hunted prey to feed their young. Evidence suggests that millions of years ago, one species of wasp started feeding their young pollen from



flowers. These

wasps were the ancestors to modern bees, who feed on nectar and pollen. In a way, you can say that bees are vegetarian wasps.

Over time, they evolved hairs on their bodies and special pollen-transporting apparatuses that allow them to collect pollen efficiently. All species of bees within the family Apidae carry pollen on their legs. Other species carry pollen inside their bodies or on the undersides of their abdomens.

(Photo credit: Jennifer Tsuruda)

Fossil records of bees preserved in amber indicate that bees have existed for at least 100 million years, but they are estimated to have first evolved around 130 million years ago, shortly after flowering plants first appeared on Earth. Plants evolved colors, odors, shapes, and food rewards that attract bees, while bees evolved mechanisms to collect pollen, which help plants reproduce through pollination. This period is known as the Cretaceous period, and was a time when dinosaurs roamed the earth. Humans did not evolve for another many million years.

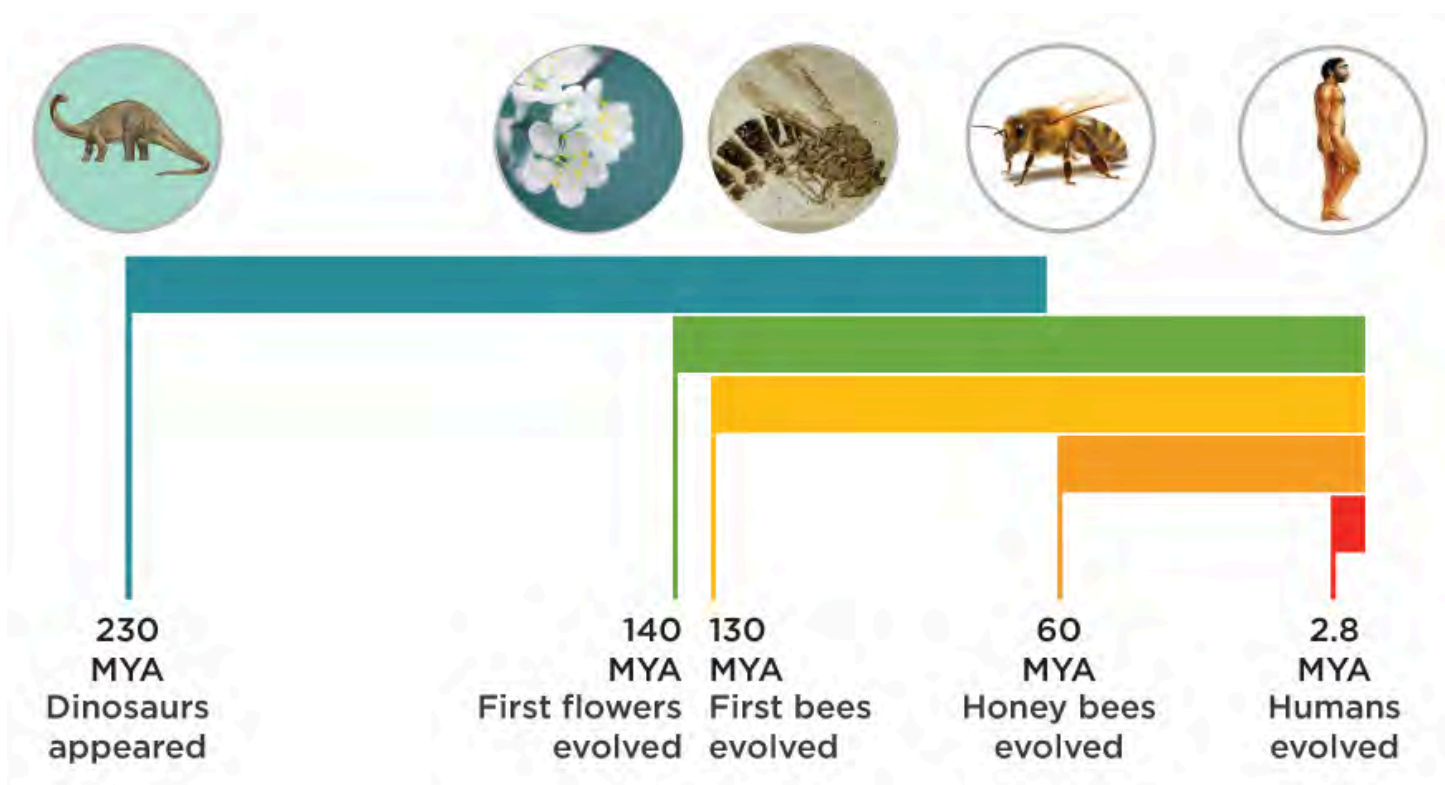
While bees have been around for roughly 130 million years, the oldest fossils of honey bees (the bees in the genus *Apis*) date back to around 60 million years ago. They occupied the so-called Old World: what is now Asia, Africa, and Europe. There is still debate over where precisely honey bees arose within the Old World. Recent genetic analyses suggest that all honey bees (genus *Apis*) actually originated in Asia, and *Apis mellifera* later diversified into Europe and Africa. They were introduced to the New World for the first time in the 1600s by European settlers. Now, they have been introduced to all continents except Antarctica.

☆ Key Points

Honey bees evolved from carnivorous, solitary wasps

Bees first evolved about 130 million years ago, shortly after flowering plants first evolved





The native habitat for *Apis mellifera* included rainforest, savannah, desert and other relatively warm climates in Africa, Asia, and southern Europe. These broad geographical ranges and differing climates allowed for several subspecies (races) of honey bees to evolve adaptations specific to their region. Honey bees have adapted to a variety of climates, including the colder temperate regions of North America. Some subspecies are better able to withstand cooler climates and store food for their year-round needs; others are better

able to avoid or combat pests and diseases that are more prevalent in warmer climates.
(Photo credit: Ben Rulison).

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Eusociality: How Honey Bees Are Truly Social

You may look at your hive and wonder how this bustling honey bee society of tens of thousands of individuals working together evolved from a solitary wasp-like organism. Honey bee colonies are examples of the most

advanced form of social living on the planet. This level of organization is called eusociality. "Eu-" means true or genuine. Individuals fully depend on a societal organization to survive and cannot live on their own.

☆ Key Points

Honey bees are eusocial, meaning that individuals depend on societal organization to survive and cannot live on their own

There are three defining characteristics of a eusocial organism:

- **Overlapping generations:** More than one generation is living together. For honey bees, there are, for the most part, two generations present in one colony: a mother (the queen) and her offspring (workers and drones).
- **Cooperative brood care:** Individuals share the duty of raising young that is not theirs. Cooperative brood care is rare in the animal kingdom. Only a few species care for their young, and the work is usually done by the mother. In the case of honey bees, the queen lays the eggs, but the larvae, pupae, and young adults are cared for by their older sisters.
- **Reproductive division of labor:** In honey bees, reproduction is divided among the members of a colony. The queen is the sole egg layer under normal conditions, and workers do not usually reproduce. The workers are not fully sterile, for they can still lay unfertilized eggs that can develop into males, but they normally do so only if they lose their queen. The three members of a bee colony (queens, workers, and drones) are divided by their reproductive ability and sex.





Honey bee colonies exhibit all three characteristics of eusociality.

Photo Credit: Bee Informed Partnership

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Watch: How Did Eusocial Societies With Distinct Castes Evolve?

Having multiple distinct forms of the same sex — like the queen and worker castes in honey bee females — is very rare in the animal kingdom. Evolutionary biologists have spent a lot of time studying other close relatives of honey bees to find out how such a caste system evolved in the first place. This video explores the prevailing theory.

Video Transcript

So how did these honey bee colonies get to have such distinct castes between queens and workers? The main theory is that the ancestor to honey bees, the solitary wasp-like organism, actually did all of the jobs that she needed to do in life by herself. She collected her own food, she built her own nest, she mated and laid her eggs and then provisioned all of those eggs with food, similar to how solitary bees live today. The resources that were available to her determined how much time and energy she was going to allocate to these different tasks. So it's not like she could be collecting food and laying eggs at the same time. There was only a short period of time where she could do each of these tasks. Importantly, once her offspring actually emerged from those eggs they left the nest. They didn't stay and hang around for any period of time to help the mother.

So at some point in the honey bee's evolution, tens of millions of years ago, there was a mutation or a genetic change that occurred. And likely, it was probably more than one. And this genetic change allowed the offspring to remain at the nest with her mother after emerging instead of flying away and starting her own nest. And this was the first time that we start to see overlapping generations present in the nest, the mother generation and the daughter generation. And we also start to see cooperative brood care, as the daughters would cooperate in rearing the brood, or their younger sisters. This change and other genetic changes were favored by natural selection and persistent in the population, allowing these family groups to start to develop, with a mother bee and helper daughter bees evolving.

So over time the mother was sort of allowed to start specializing in reproduction and laying eggs, while the daughters, this kind of frees up time for them to start specializing in other helping strategies. Over time natural selection sort of continued to favor this and then these rigid castes were finally in place. Their evolved, there's reproductive division of labor between the queen and reproduction and the workers in other tasks. In the case of



honey bees, these castes have become so rigid that female bees must function either as a queen or as a worker. It's not like a queen bee can revert back to being a worker and vice versa. A worker can't psychologically change to become a queen. Some other species of bees like bumble bees aren't quite as distinct and rigid. They still have queens and workers, but if a queen suddenly dies, a large worker can actually start to dominate her sisters and it is physiologically possible for her to mate and lay eggs. Where as it is not physiologically possible for workers to mate.

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Watch: Kin Selection

The evolution of a non-reproducing worker caste is one of the most extreme examples of altruism in nature, and has fascinated biologists for centuries. The term 'altruism' refers to the fact that the majority of female bees in a colony suppress their own reproduction and instead invest energy into helping another individual (their mother, the queen) reproduce. On the surface, this appears to go against nature. Reproducing to pass its genes on to the next generation is, biologically speaking, the most important thing an organism can do – it is usually the measure of success for living things.

Charles Darwin, who first proposed the theory of evolution, wondered how individuals who refrain from reproducing could evolve. If an organism does not have any offspring, how do its genes persist in the world? And why would some bees refrain from reproducing in order to raise another bee's young instead? In *The Origin of Species* (1859), he describes the non-reproducing caste of honey bees as “one special difficulty, which at first appeared to me insuperable, and actually fatal to my theory”. It was not until the 1960s that a clear explanation arose: kin selection. Watch the video below to learn about kin selection:

Video Transcript

Workers inhibit their own reproduction in order to help the queen reproduce. And it's really costly for them to invest so much energy into rearing the queen's offspring. It's costly for them to go out and collect pollen from flowers, and risky because they might come in contact with predators. It's costly for them to produce royal jelly in their glands and to feed that to the offspring and to keep the brood warm. It's also costly for them to keep on checking on the brood and making sure that they're developing okay and that they're not getting any diseases. And most of all, one of the biggest costs is that they sacrifice their own lives to sting intruders to protect the colony. So all these things are really energetically expensive. So why are workers behaving so selfless? The answer is kin selection. Even in humans, the want to help is more likely with kin or your relatives than non-kin. Think of how protective you feel if someone wants to fight one of your siblings. Or a time when you might put yourself in danger or in a risky situation to try to save one of your relatives. Another good example of family helping is when there's a new baby in the family and aunts and nieces and other relatives will come together and help raise the baby.



Kin selection is a strategy where individuals reduce their own reproductive success or survival in order to increase the reproductive success of someone else. And in the honey bee colony it's the worker who's reducing their reproduction to help their mom, the queen, reproduce. On the surface this appears really selfless, but at the genetic level you can start to see why it's advantageous and why this behavior likely evolved. Being related means you share genes with someone and the amount of genes that you share influences your helping behavior. And humans and bees are different. They differ in how they inherit their genes and how related they are with their kin. So let's start with humans. Individuals inherit genes from their parents, so you have genes in common with all your relatives: your parents, your grandparents, your siblings and your own children. And the closer you are related, the more genes you're actually sharing. So you share more genes with your mom than you would with your great, great grandmother, for example. Humans are diploid organisms. This means that they have two sets of each chromosome in all of their cells. So their skin cells, their bone cells, their muscle cells, they all have two sets of chromosomes. One set is inherited from your mom, and one set is inherited from your dad. The exception is when you're looking at your gametes. Your egg cells or sperm cells only have one set of chromosomes. So because you're inheriting one set from your dad, and one set from your mom, this means that 50% of your genes are shared with your mom, and 50% of your genes are shared with your dad. And also as a result, 50% of your genes are shared with your sibling. Looking at a family tree, we can see the average degree of genetic relatedness. And this just goes to show again how many genes you're sharing. So, mothers and daughters share 50%, fathers and daughters share 50%. And same with your brother, they share 50% with their mom and 50% with their dad.

Honey bees have a slightly different genetic system and it's called haplodiploidy. The females are similar to humans where they come from a fertilized egg so a sperm and an egg come together and they have two set of chromosomes. So they're also diploid. But, the males are different. Instead, males come from unfertilized eggs. They only inherit one set of chromosomes. That from their mother. So they are called haploid. They have half the number of chromosomes than the females in the colony do. So drones are haploids and all the females, workers and queens are diploid. A haplodiploid system actually makes bees more related than 50%. So this is where we see the difference from humans. In fact, if a queen only mates with one male, workers will share 75% of their genes. So how does that even work? Well, both drones and queens are passing on one copy of their DNA to create diploid daughters. And since the drones only have one set of chromosomes to begin with, all the copies that he's passing on to each daughter are going to be identical.



So, sisters are sharing 100% of genes that they receive from their father. But with the queen she's diploid. She has two sets of each copy of each gene in her DNA. And she gives one copy to each of her daughters. So one sister might inherit the same set of chromosomes or some of the same genes as the other sister. But sometimes they receive different copies. Across the thousands of genes that they're inheriting the amount of times they end up sharing the same copies averages out to about 50%. It's similar to tossing a coin. There's two options. You can have heads or you can have tails. If you and your sister are each flipping the coin over time, over thousands of tries, the average amount of time that you are going to get the same copies is about 50%. So sisters share about 50% of their genes that they receive from their mother. So a daughter of a singly mated queen who has the same dad will share on average 75% of their genes; 100% from the dad side and 50% from the mom side averages out to 75%.

Okay, so what if a worker were to lay her own eggs? If a worker were to lay her own egg she would pass on 50% of her DNA. Workers can't mate or store any sperm so they're producing unfertilized haploid drones. They're always drones. So if we're thinking about it, what is the better avenue to get the most genes to persist in the population? Is it to have your own offspring, where you're passing on 50% of your genes? Or is it to help rear your sisters, who are going to be sharing 75% of your genes? It's to help raise your sisters. And if we are looking at a honey bee colony, we see that the queen is able to lay many many eggs a day, up to a thousand eggs a day, so it is a really good way to pass on your genetic information even though you yourself aren't reproducing or the workers themselves are not reproducing.

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Read: Polyandry

When honey bees first evolved, queens were monogamous, meaning they mated with just one male. This meant that all of her daughter bees were highly related to one another and had 75% of their genes in common. This high level of relatedness helped drive the evolution of eusociality.

Once eusociality was established, however, other colony traits also evolved. One such trait is polyandry. Polyandry is the practice of females mating with multiple (poly) males (andry). Queen honey bees are promiscuous, typically mating with between 12 and 50 males. By mating with multiple males, a queen greatly increases the genetic diversity among her daughters, the workers in the colony, thus improving the colony's health and productivity.

☆ Key Points

Queen honey bees usually mate with between 12 and 50 males, which helps to increase genetic diversity among her offspring, the colony's workers.

Genetic diversity improves colony health and productivity.

Relatedness among sisters decreases with polyandry, but it is still more advantageous for workers to rear the queen's offspring (their brothers and sisters) rather than their fellow sisters' offspring (their nephews).



Queens mate with multiple drones.

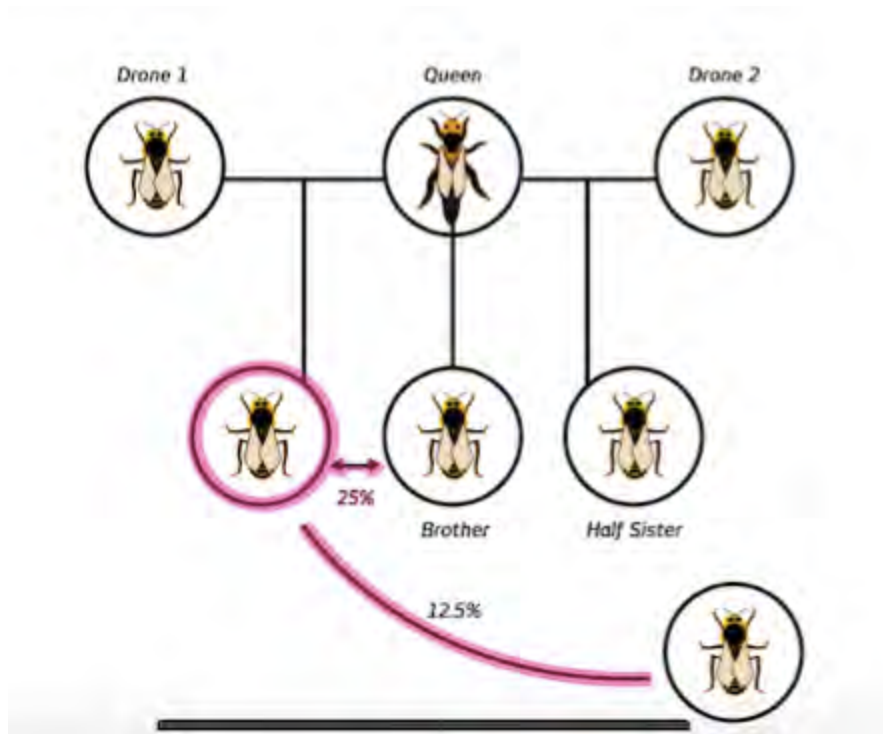
The Impact of Polyandry on Relatedness

When queens mate with multiple males, the relatedness between sisters decreases. While they share 75% of their genes when they have the same father, daughter bees only share 25% of their genes when they have unrelated fathers. Under these circumstances, one might expect workers would produce their own offspring instead of helping rear their sisters. Yet workers still refrain from reproducing, despite the fact that they would now be more related to their offspring than their half-sisters. How come?

One possible reason has to do with the high reproductive output of the queen. The queen can lay 1,000 to 1,500 eggs per day, while a laying worker lays only a few eggs per day. Even though workers are less related to the queen's offspring than their own, they can still effectively pass on more of their genes by helping the queen.

Another possible reason has to do with the relatedness between a worker and her sisters' offspring. Although it would be advantageous for a worker to rear her own offspring, it would not be advantageous for her to rear her sisters' offspring. If several workers began laying eggs in the colony, one worker would invest energy in rearing many of her nephews as well. In this situation, it is more advantageous for sisters to invest energy and resources into rearing their mother's offspring (with whom they share 25% of their genes) than their sisters' offspring — their nephews — with whom they only share 12.5% of their genes.





A specific social behavior has evolved to discourage workers from reproducing. This behavior is called **worker policing**, and it refers to workers consuming the eggs that are laid by their sisters. In the rare instances where workers do lay eggs, their sisters will eat them so that they never develop.

We've talked about the genetic relatedness among several family members in a honey bee colony, and it is possible to calculate these values for all their relatives. It is, however, beyond the scope of this course to have you calculate genetic relatedness among family members. The take-home message is this: Understanding genetic relatedness among family members in a bee colony can help us learn how and why honey bees evolved such a social living strategy.

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Watch: The Importance of Wild Bees

There is incredible diversity in bees and striking differences in the way they look, behave, build their nests, and live alone or together. Although honey

bees are the species most commonly used for pollination in agriculture, many species of bees also have important roles in pollinating food crops and wildflowers. Across the world, there are over 20,000 documented species of bees! Watch this video to hear Bryan Danforth discuss the many other bees (besides honey bees) that are found in the world.

☆ Key Points

There are 20,000 species of wild bees in the world

Video Transcript

So honey bees are not the only bees in the world. In fact, there are over 20,000 described species of bees in the world. In North America, we estimate about 4,000 described bees. California hosts over 1,000 species of bees and French-speaking Europe, hosts nearly a 1,000 species of bees as well. In New York state, we have recently tallied up the numbers and come up with a total 420 species. So, while these are extraordinarily diverse, and if you compare the species richness of wild bees to other familiar large vertebrate groups, bees are comparable to fishes in the vertebrates with about 20,000 species. There are about five times as many bees as there are mammals and there are about three times as many bees as there are birds. So bees are an enormously diverse group. They're a group that arose over 120 million years ago. That's well before the extinction of the dinosaurs and it's a period when the earliest flowering plants were just evolving on earth.

I'll use two terms throughout this presentation. One is native bee to describe bees that are native to a region or of the world. And wild bee to describe all the non-managed bees that we'll be covering in today's presentation. Bees occur on all continents except Antarctica and they're particularly diverse in desert habitats such as the circa Mediterranean region, western Australia, Southern Africa, and western North America. By comparison, honey bees encompass only eight species in the genus *Apis*. And one of those, the managed honey bee, is *Apis mellifera*. Now it turns out wild bees are actually very important agricultural pollinators and we've learned this fairly recently, from studies in crops like apples, peaches, cherries, blueberries, cranberries, strawberries, squashes and pumpkins. Wild bees are a large component of the fauna in these agricultural habitats. And in New York, for example, we have 120 species of wild bees visiting apple trees, just alone. Apple



growers, for example, in New York and other agricultural producers are well aware of the role of these wild bees. And they do a very good job of trying to maintain a diverse wild bee fauna in and around their cultural habitats.

Native bees and the wild bees we're talking about today have coevolved with many different host plants. And one of the interesting things about native wild bees is that they are often narrow host plant specialists, unlike the honey bee, which is a very broad generalist visiting many different host plant families. So what do we know about these other wild bee species? First of all, they are extraordinarily diverse in anatomy and behavior. There are a tiny species the size of a fruit fly in arid regions of western North America and there are enormous carpenter bees the size of a bird that one can see flying in, more or less, tropical habitats. Bees come in a range of colors and in different amounts of hairiness. But they're quite diverse anatomically and behaviorally. The vast majority of bees are solitary. So our estimate is that about 77% of the bee species on earth are solitary. A single female constructs her own nest. She guards that nest against parasites and predators. She gathers pollen and nectar for her offspring, and she lays eggs. She's usually active for a very short period of time. Somewhere between two to three weeks. In contrast to a honey bee colony which is active for months throughout the summer. And again, many of these solitary wild bees are narrow host plants specialists, visiting just a few host plant genera or even species.

Now these wild bees build nests in a variety of substrates, including soil, hollow stems, wood, snail shells, some of them build above ground nests, and some of them build nests in solid wood, those are the carpenter bees. So let's start with the soil nesters which are the largest proportion of wild bees on earth. These are bees that burrow into the soil. They can build nests from just a few inches deep to up to nine feet deep. There are records of these bees building nests that go down nine feet into the soil. At the bottom of their nest, they will construct a series of underground brood chambers which they provision with pollen and they'll waterproof these brood chambers with glandular secretions or floral oils. And then they'll provision that brood chamber by making many trips, usually five to ten foraging trips. They'll lay an egg on that collected pollen and then they'll close that brood chamber and start the whole cycle again. A typical solitary bee produces about one offspring per day, but some very rapid foragers can do five to even six offspring per day. And an average wild bee doesn't live for very long. She might produce 10, 12, 15 offspring over the course of her entire life.



Another group of solitary bees are the cavity and stem nesters. These include the mason bees, the leaf cutter bees, and the wool carder bees. And we call these renters because they occupy cavities that were created by another organism or just cavities available in the environment. And what they do is they construct their brood cells within these cavities by modifying the cavity slightly with mud, leaves, plant fibers, even flower petals and once they've constructed their brood chamber with those materials they then provision the brood chamber with pollen and nectar just like the ground nesting bee would do. They lay an egg and then they close the brood chamber and carry on with another brood cell the next day. Another group of wild bees are the wooden pith-nesters. And these include the large carpenter bees as well as the small carpenter bees. These are bees that construct a burrow in solid wood or in pithy stems. And carpenter bees have an enormous head with stout mandibles for burrowing into solid wood and constructing a series of burrows. and then from within those burrows they construct their brood cells. They provision their brood cells again with pollen and nectar, they lay an egg and they close the brood cell. And they will construct multiple brood cells throughout their lives.

And finally, the last group of wild bees are the resin bees and these are bees that actually construct above-ground nests from resins combined with pieces of stone or chewed up leaves. Again, they use a variety of floral products, but they actually construct their nest above-ground, and they attach it to a plant or a stone, almost like a little flower pot. And then they forage out from that nest collecting pollen and nectar, and they bring those floral resources back to their nest, they provision their brood cells, they lay an egg, and they close that brood cell off. So those are the resin bees in the family megachilidae. And again these are all solitary, like most of the bees I've told you about so far.

Now one of the really fascinating groups of bees are the kleptoparasites. These are bees that very few people know about. They comprise 13% of wild bee species, and almost 25% of the family Apidae which is the group that includes the honey bee. These are bees that don't construct a nest. They don't gather pollen and nectar. They sneak into the nest of other bees. And they lay an egg discretely in the brood cell of that host and then they leave the nest as quickly as possible so as not to alert the host to their presence. Their egg hatches and the first instar larva usually seeks out the host egg or early instar larva and they kill it with greatly enlarged mandibles. And, then the klepto parasite larva consumes the pollen provisions gathered by the host female. So these are really similar to the cuckoo birds that some of you might be familiar with that are known for birds where the cuckoo lays eggs in another birds nest and then flies away and hopes that that bird will take care of its offspring. So these are called cuckoo bees or kleptoparasites or brood parasites. And



you can often see them if you look around bee nesting sites, if you look near the ground, you'll see what look like wasps almost flying near the ground and looking for nests of their hosts and then entering those nests to lay their eggs. So, we don't know whether these are good pollinators but they're certainly fascinating bees that have adopted a very unusual lifestyle.

And then the last category of bees are the Social Bees, which of course would include the honey bee. In this group only includes 9% of the bees in the world. Social bees other than honey bee would include bumble bees, sweat bees and stingless bees in the tropical parts of the world. Stingless bees are comparable to honey bees in terms of social complexity. They build enormous colonies with thousands of workers. They are aggressive defenders of their nest. They communicate about the location of resources. They've actually been managed in various parts of the world for honey production, just like we manage honey bees today in North America. It's a little more difficult to manage a stainless bee colony for honey production, so they're not as widely managed as the honey bee. So while honey bees are fascinating and extremely effective pollinators in all kinds of crop systems, we shouldn't forget that wild bees are important pollinators, but also fascinating from a biological perspective. And what we're discovering with our work on wild bees of apples is that conserving wild bees is an important approach for agricultural producers who want to maximize the pollination of their crops. It's also important for environmental conservation and for carrying on sustainable agricultural practices.

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Read: Conserving Wild Bees

There is a lot of interest today in bee conservation. Honey bees are not alone in facing impacts from parasites, diseases, nutrition, and pesticides; some wild bee populations are also facing challenges from these factors.

Their conservation is equally important for maintaining natural ecosystems and sustainable agriculture. Wild bees contribute to overall ecosystem health and the biodiversity of native flowers, and as beekeepers, we are regarded as stewards of all pollinators and advocates for healthy environments. Although honey bees comprise only 9% of bee species (and aren't native to North America), they have become the face of pollinator declines in the media. Master Beekeepers can use this opportunity to help educate the wider public about all kinds of bees and pollinators. The beekeeper's voice is an important one.

In the Northeast U.S., there are 53 species of wild bees whose populations are currently in decline. In 2016, several species of bees were added to the Endangered Species list for the first time in history. These bees were seven species of yellow-faced bees that nest in small cavities and stems in Hawaii. In early 2017, the rusty patched bumble bee, native to eastern North America, was also added to the list.

With most research focused on the factors affecting the health of honey bees, we are only recently identifying how wild bees are impacted by some of the same factors. Wild bees, for instance, are more susceptible to habitat loss than are honey bees. While wildflowers generally only provide food for honey bees in the U.S., they also provide nesting sites for some native bees. Habitat loss in the form of agricultural intensification and urbanization has been identified as one of the main driving factors for reducing some wild bee populations.

The impact of various pesticides on wild bees is only beginning to be investigated. We are learning that pesticides affect different bee species differently — some are much more sensitive to certain chemicals than honey bees. We are also starting to better understand the role of certain parasites and pathogens in wild bee health. At least two are shared between honey bees and bumble bees; *Nosema ceranae* and Deformed Wing Virus can

☆ Key Points

Some wild bee species are experiencing population declines, and several have been added to the Endangered Species List



infect both genera ('genera' is the plural of genus). Research on how pesticides and pathogens are impacting native bee populations is currently underway at Cornell and other institutions.

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Quiz: The Evolution of the Honey Bee Colony as a Superorganism

You must achieve a score of 100% on this quiz to complete the course. You may take the quiz as many times as you need to achieve that score.

Please complete this quiz in the course.

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Discussion: Module 1 Discussion and Questions

This course provides the opportunity to discuss content with fellow beekeepers in addition to the course instructor. You are welcome to participate in all course discussions but are only *required* to participate in two specific discussions (Module 2 and Module 6.)

This Module 1 discussion is **optional** and your participation in it does not count toward your final grade. We encourage you to read other students' comments and to respond if you are interested.

1. Please answer the following questions about **Module 1: The Evolution of the Honey Bee Colony as a Superorganism**:
 - Throughout history, honey bees have been used as a metaphor and model for human society. Which aspect of their sociality do you think is most interesting, surprising, or important?
 - What questions do you have for the instructor related to the information provided in this module?
2. After you have completed your original post, take a moment to review and comment on the thoughts and ideas posted by others. As you do so, consider the following:
 - What aspects of your classmates' observations about honey bee sociality resonate with your own views?
 - What different perspectives did they offer that you found interesting?
 - Based on your own experience or knowledge, what additional insights can you add to your peers' observations?
 - What questions or thoughts did your peers' posts raise for you about honey bee colony behavior and organization?
 - How do your peers' interpretations of the honey bee colony as a metaphor for human society compare to your own views? What new angles did they present?

To participate in this discussion:

Click **Reply** to post a comment or reply to another comment. Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [eCornell's policy regarding plagiarism](#) (the presentation of someone else's work as your own without source credit).

Please note:



While discussion board postings will be accepted through the end of the course, we strongly encourage everyone to move through this course as a group. As such, postings that are made to this board after 5pm ET on the due date will be read but may not receive a response on the board from your facilitator and/or your peers. Please let your facilitator know if you have any questions.

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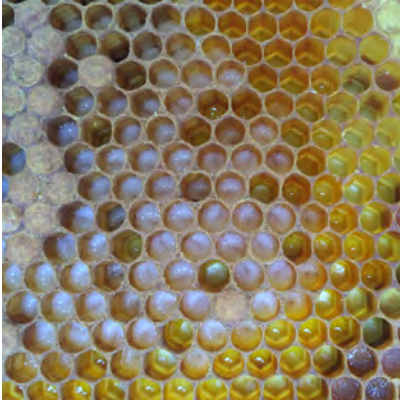
Module Wrap-up: The Evolution of the Honey Bee Colony as a Superorganism

In this module, you explored the remarkable evolution of honey bees from solitary wasps to one of nature's most sophisticated social organisms. Understanding how individual bees function as part of a larger, interconnected colony helps beekeepers appreciate the complex adaptations that make honey bees such successful and fascinating creatures.

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Module Introduction: **Honey Bee Reproduction**



Reproduction in honey bee colonies occurs at two levels:

- The entire colony can reproduce through swarming.
- Individuals within a colony can reproduce by mating and having offspring.

This module focuses on *individual* reproduction and covers everything you need to know about the biology and behaviors involved.

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Read: Passing on Your Genes

Through reproduction, bees pass on their genetic material to their offspring. This course uses a variety of terms when talking about genetic material: genome, chromosome, gene, and DNA. These words are all related, but mean slightly different things. Let's get familiar with these terms so that we can better understand honey bee reproduction and genetic characteristics.

Basic Genetics

A genome is an organism's complete set of genetic material.

Deoxyribonucleic acid (DNA) is the name of the molecule that carries this genetic material. The honey bee genome is different from the human genome and the genomes of all other organisms because each species has differences in their DNA that make them unique. The genome includes information for a variety of elements, the most important being genes. The honey bee genome was sequenced in 2006 and is one-tenth of the size of the human genome.

An organism's DNA strands are incredibly long. For instance, if a human's DNA was stretched out it would be three meters in length! In order to fit all the DNA into each cell of an organism's body, it is tightly coiled up into molecules called chromosomes. Honey bee DNA is distributed across 16 different chromosomes:

- **Queens and workers.** Diploid females carry two sets of chromosomes, so each cell (except their egg cells) contains 32 chromosomes total.
- **Drones.** Haploid males carry only one set of chromosomes, so each cell (including their sperm cells) contains 16 chromosomes total.

An organism's DNA serves as the blueprint for its body. Different regions of the DNA strands are the written instructions for different genes. The honey bee genome contains

☆ Key Points

DNA is the name of the molecule that carries genetic material. An organism's DNA is coiled up into chromosomes

Genes are sections of DNA that code for particular proteins. Genes are passed down to offspring through reproduction

Beekeepers can selectively breed for the characteristics they desire

Many honey bee behaviors are governed by multiple genes



around 10,000 different genes, so each chromosome contains the instructions for many hundreds of different genes. Small particles within cells read the gene instructions and build their corresponding proteins. For example, a gene with instructions to build an eye cell would get translated into an eye cell protein.

Like most other animals, honey bees reproduce by creating egg cells and sperm cells. These cells contain the parents' chromosomes, so all offspring that are produced inherit their parents' genes.

How would a beekeeper breed for the trait they desire?

Many of the traits we observe in honey bees are due at least in part to genes, so beekeepers can selectively breed for characteristics they desire. For example, there are many genes involved in nectar collecting, in temperament, and in hygienic behavior. Bee breeders can try to improve their operation's honey production, gentleness, or colony health by breeding queens from colonies that have these traits. Honey bee behaviors are complex because they tend to be governed by multiple genes and the environment also influences their behaviors, but nonetheless beekeepers have been breeding for traits like these for decades - long before we knew about DNA.

All honey bees are recognizably honey bees, but no two are identical. While they all contain 10,000 or so genes in their chromosomes, the *versions* of those genes can be different. To illustrate this point, let's think of an example in humans: eye color. Everyone has a gene that codes for eye color located on the same position on one of their chromosomes, but people can have different colors of eyes: some have brown, while others have blue or green. This is due to different versions of the eye color gene. Individuals inherit their specific gene versions from their parents. Scientists refer to different gene versions as "alleles".

Now let's look at an example with honey bees: temperament. Let's imagine that we have a bee yard with many hives in it, and the colonies vary greatly in their temperament. Some colonies are very gentle, while others are much more prone to sting. In the honey bee genome, there is one gene that codes for temperament (in reality temperament is governed by many different genes, but we will keep this example simple). Some colonies carry a version of this gene that makes them more aggressive, while other colonies carry a version of this gene that makes them gentle. Over the years, we can improve the temperament of our entire bee yard by selecting which gene versions are going to be passed down to subsequent generations. We can allow the queens from gentle colonies to reproduce, and can prevent the queens from aggressive colonies from reproducing (by



replacing them with queens raised from our gentle colonies). Over a few bee generations, we can have a yard of entirely gentle colonies, as that gene version becomes predominant over the other version in the population.

By understanding honey bee reproduction in this way, beekeepers can start to breed for characteristics they desire - whether their goal is to increase honey production or to help improve honey bee health in an effort to be a better steward to the environment.

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Read: The Molecules of Bee Life

DNA and chromosomes are important molecules of bee life, but there are others to consider as well. We don't often think of tiny microscopic molecules when we think of honey bee reproduction, development, and growth, but in order to understand how these processes occur we need to take a step deeper than what we see with the naked eye. It's interesting to know that honey bees reproduce and develop in unique ways as animals, but it is even more valuable to understand *how* they do it.

Chemical compounds in the form of molecules are key players in development and reproduction of all living organisms, for they enable communication among the cells inside a body and among the individuals within a honey bee colony. The molecules important to honey bee reproduction, development, and sociality that we will cover in this course are proteins, neurotransmitters, hormones, and pheromones.

▼ Proteins

Proteins form the building blocks of life (human, honey bee, and every other living thing) because they are involved in so many functions. For example, many organs and tissues (e.g. hearts and muscles) are made primarily of different types of proteins. Even the immune system relies on proteins to function. Proteins also play a big role in development and reproduction. The instructions for building proteins are written in DNA (deoxyribonucleic acid), found in most cells within an organism.

▼ Neurotransmitters

Neurotransmitters are chemicals released by neurons in the brain to send very fast signals to other neurons. They are the important messenger molecules of the nervous system and are involved in many functions ranging from muscle movements to learning and memory in the bee. Many neurotransmitters in honey bees are also present in humans, such as dopamine, octopamine, and serotonin.



▼ Hormones

Hormones are chemicals produced in special glands that are secreted into the blood (called the hemolymph, in bees) to be carried to specific organs and tissues to influence how these structures function. Hormones travel within the body and do not exit it. In the honey bee, as in humans, hormones are involved in reproduction, development, and the general functioning of the body (e.g., stress hormones).

▼ Pheromones

Pheromones, like hormones, are also chemicals created by specialized glands, but they are secreted outside of the body to elicit a behavioral or physiological response by another individual of the same species. Insects, vertebrates (including humans), plants, and even yeast commonly use pheromones as a means of communication.



Some pheromones function as airborne molecules (odors) that are detected through the sense of smell, while others function as surface-borne molecules that are secreted onto the outside of the body where they can be detected through taste or close-range smell.

Some honey bee pheromones are even transmitted throughout the colony by worker bees engaging in trophallaxis (transferring or exchanging fluids or food mouth-to-mouth).

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Watch: Mating

In order to pass their genes on to their offspring, honey bees mate. Queens and drones are the only colony members that mate. Mating activity in honey bee populations typically follows a seasonal cycle that coincides with swarm season. When colonies swarm, they leave behind new virgin queens. Drones are produced amply in spring and summer to mate with these queens.

In northeast North America, this mating activity peaks in the spring and early summer but continues to occur to a lesser degree throughout the summer as queens are superseded and late swarms occur. Watch this video for details of where and how mating happens.

Video Transcript

So when a virgin queen is sexually mature, she's ready to take a mating flight and she'll take between one and three mating flights where she leaves the hive and flies to the specific geographical location called the drone congregation area. And this is an area high in the sky. It's between 50 to 250 feet up and it's just essentially a cloud with hundreds of drones that are waiting for a queen to arrive. So queens will fly up to nine miles to get to these drone congregation areas. Even if there's one near her own hive, she wants to fly far. And this is so that she reduces the chances of mating with her brothers and other relatives. And it's actually quite effective. Only a very small percentage of queens are mating with drones from the same apiary.

So drones actually get a little bit of a bad rep. People often say that drones are lazy and they don't do too much, and the word drone is literally synonymous with the word lazy. But when it comes to reproduction, drones are anything but lazy. What they do is they leave their colony many times a day to fly to this drone congregation area, where they hang around for about half an hour waiting for a queen to show up. After about a half an hour, they're very hungry, so they have to return to the hive refuel, and they set out again. So they do this several times, just hopelessly waiting for a queen and seeing if they're going to get lucky. So a really cool thing about drone congregation areas is that queens and drones from colonies in a given region actually return to the same drone congregation area year after year. In fact, right here next to Dyce, we have a drone congregation area that people have observed for decades. But we really have no idea what attracts drones and queens to these congregation areas to begin with, or how they



become established, or how they know to continue going to these spots year after year. We do know that there are some traits that make drone congregation areas more attractive. So for instance, open areas are more attractive than areas that are near dense forests or urban areas and congregation areas that are above slopes that are facing southward are more attractive than slopes facing any other direction. But beyond that, really a lot more research needs to be done and drone congregation areas still have a lot of mystery to us.

Okay, so let's talk about the act of mating. We have hundreds of drones in these drone congregation areas and suddenly a queen arrives, this is the moment they've been waiting for. And the drones can detect the queen with two ways. First is sight, they have very large eyes. They have to be able to distinguish one queen in a cloud of hundreds of drones. And the second is smell. So queens are continuously producing a combination of pheromones that are collectively called queen substance or queen pheromone. And it acts as a sex pheromone to males. It's very attractive, and it makes them want to mate with her. So once they see her or smell her, the race is on. And they are trying to get to the queen. And out of hundreds of drones in a drone congregation area she only mates with between 10 and 20 other males. So the competition is fierce. The actual endeavor of mating is very quick, it lasts less than five seconds. Normally the drone's penis is tucked underneath its body, but it gets everted when it's ready to mate, and inserted in the queen's vagina. Less than five seconds later, he ejaculates, and this actually ruptures his penis. His body falls away and he dies, and the tip of the penis remains in the queen. The next drone to meet with the queen will remove the tip with this specialized structure on his own penis, and then he will mate with her. So consecutively she'll mate with 10 to 20 drones and then when she's finally finished mating with all the drones she's going to, she returns to the hive and the last tip remains. And some beekeepers actually witness this and they can see this, and it's called the mating sign. And it says to them that they have a mated queen. So it's actually a good thing if you happen to see this. The workers in the colony will remove the last tip.

So after the queen returns to the hive from her one to three mating flights she'll never mate again. All the sperm that's stored inside of her is the sperm she's going to use to fertilize all of her eggs. And it's going to be stored inside of her for several years. You can tell that she's mated because her abdomen is larger, it's more full of sperm. She's actually moving slower in the colony because she's heavier, and she won't fly again unless she's going to swarm. After she's returned from a mating flight, two to four days later she'll begin laying eggs, and this is what she continues to do for the rest of her life.



Most mating flights take place in the afternoon, usually between 2:00 and 5:00 p.m. While queens can fly quite far to drone congregation areas (DCAs), on average they will typically fly 1.2 to 3 miles (2 to 5 km). Drones, on the other hand, will visit much closer DCAs that average 0.3 to 1.2 miles (0.5 to 2 km). If drones do not get to mate with a virgin queen on the first flight they take, they will continue to take up to six 30-minute flights each day as the weather permits. In between flights, they return to the hive to refuel on carbohydrates.

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Read: Reproductive Systems

The physiology of reproduction varies by sex and by caste. Review the information on each tab below to understand the reproductive physiology of the drone, the queen, and the worker bee, as well as the role each plays in the process of reproduction.

☆ Key Points

Queens have a very high reproductive output and signal their fertility to the colony through queen pheromone

Worker honey bees have similar reproductive systems to queens, but they have fewer ovarioles and are not able to mate

Drone

The Reproductive System of a Drone

Drones become sexually mature and ready to mate after nine to 12 days of adult life. The drone's penis (called an endophallus) only everts when he latches onto a queen to mate; it is normally tucked inside the drone's abdomen and is not visible.

Because the bees are mating while flying, the drone's endophallus also contains a pair of claspers for gripping onto the queen. In a successful mating, 6 to 12 million sperm travel through the endophallus and into the queen's vagina. Many fewer sperm than this make it to a queen's sperm storage sac (the spermatheca); the rest exit her body.

Queen



The Reproductive System of the Queen

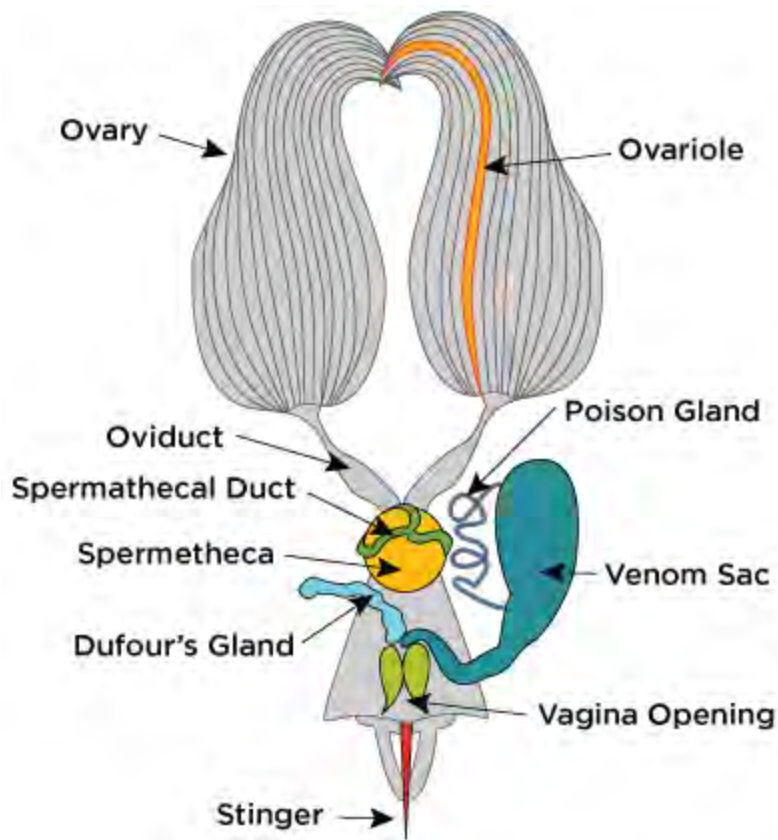
The queen bee is an egg-laying machine for nearly her entire lifetime: In the right conditions, she lays about one egg every minute, which translates to 1,000 to 1,500 eggs a day and about a half-million eggs in her two- to three-year lifespan, although this can vary enormously. Some sources have stated that some queens can lay over 2,000 eggs a day!



When the quality or quantity of her egg laying declines and signs of her fertility weaken, the workers supersede her with a new queen. *Queen eggs laid in cells.*
Photo credit: Bee Informed Partnership.

The ovaries in queen bees are two large, tube-like organs that are comprised of 150 to 180 ovarioles each. All of the queen's eggs are produced in these ovarioles. When an egg is released, it leaves an ovariole, slides down an oviduct, passes by the spermatheca, moves through the vagina, and exits the queen's body from her vagina. The spermatheca is a large sac that stores sperm from all the drones a queen has mated with; about 5 to 6 million sperm total. To fertilize an egg, a small duct at the entrance of the spermatheca opens as the egg is passing by to allow sperm to exit and fertilize the egg. In this way, queens are able to control whether they lay fertilized eggs that will develop into workers or unfertilized eggs that will develop into drones. There is generally enough sperm in a spermatheca to last for a few years of egg laying; only a few sperm are released to fertilize each egg. Note that a queen who has just mated may, for a time, lay multiple eggs into a cell, but





within a couple of weeks, she should settle into a productive and sustainable pattern of egg laying.

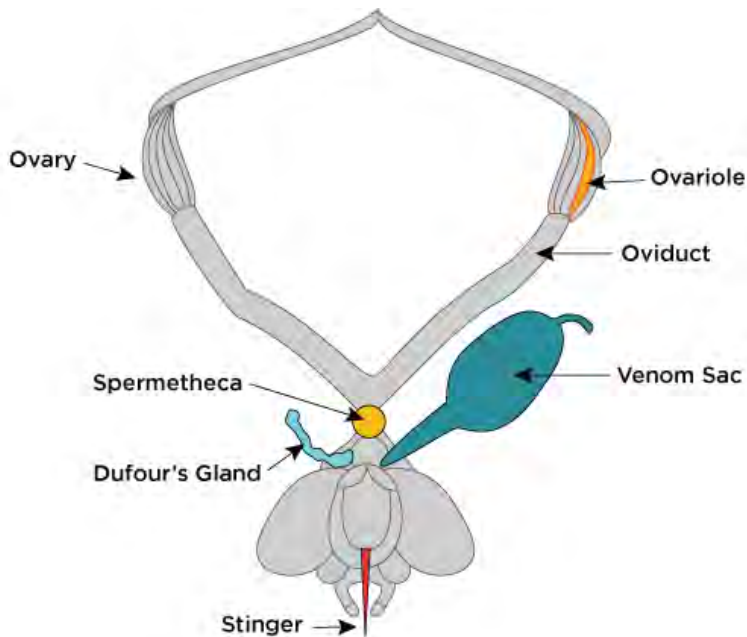
The queen (marked) signals her fertility by producing a blend of pheromones, collectively called "queen pheromone." Workers pass along this information about her fertility to other bees in the colony by transmitting

the pheromone to them. Workers near the queen brush this pheromone blend off the queen's body with their antennae and lick it off with their tongues, then walk through the hive spreading it to one another by making antennal contacts and via trophallaxis. In this way, they can constantly assess and communicate the presence and fertility of the queen. As the queen ages and her fertility wanes, she produces less queen pheromone. Once worker bees can no longer detect queen pheromone at a minimum threshold, they start to construct supersedure cells to replace her. If the queen dies or is removed from the hive by a beekeeper, the workers come to realize she is gone because of the lack of queen pheromone being passed among the bees. In response, they begin constructing emergency queen cells to rear a new queen.



Worker

The Reproductive System of the Worker



While queens and workers have the same basic biological equipment for reproduction, they differ vastly in their ability to lay eggs. Under normal circumstances, when the hive is queenright, worker bees are effectively sterile. They have two ovaries just as queens do, but these ovaries are small, with only two to 12 ovarioles each, in striking contrast to the 150 to 180 ovarioles in each ovary of a queen. Similarly, a worker's spermatheca is tiny and cannot

store sperm. Because of this, workers cannot mate and do not tend to lay eggs. In circumstances when they do lay eggs, their eggs are unfertilized and develop into haploid drones.

There can be a few laying workers even in a queenright hive, but rarely do they make up more than 1% of the worker population, and the eggs that they lay are usually eaten by other workers. If the queen dies, workers can detect the absence of queen mandibular pheromone in as little as 20 minutes. If the colony is not successful in rearing a new queen from emergency queen cells, some of the workers' ovaries may activate, and they can start laying eggs, becoming what are called "laying workers."

The young workers (nurse bees) in particular are affected by shifts in this pheromone, as they are much more likely than the foragers to be in the hive and in



contact with the queen. It is these young house bees that can become reproductive. In the absence of queen pheromone, up to 30% of young workers in the colony may start to produce and lay eggs! The reproductive output of laying workers is very different from queens. Laying workers only lay a few eggs here and there — not nearly the 1,000 or so per day that a queen is able to lay.

Because workers have short abdomens that cannot reach the bottom of a cell, their eggs tend to be on the sides of cells rather than at the bottom. Also, there is often more than one egg per cell. Beekeepers often look for the eggs to confirm the presence of the queen — but Master Beekeepers must be able to inspect the egg pattern and identify whether the eggs were laid by workers or by a queen. Workers will often lay eggs in worker cells, but because they cannot mate, all the eggs they lay will develop into drones. The drones' larger bodies will cause the cell cappings to extend outward and become bullet-shaped. These bullet-shaped cells, in combination with multiple eggs on the side of cells, is a good indicator that there are laying workers present in the colony.



Worker eggs in cells look like several small white eggs crowded together.





Worker eggs laid in worker cells develop into drones. Because drones are larger, they form "bullet"-shaped cells when they are capped.

Photo Credit: Bee Informed Partnership

Finally, an important part of reproductive anatomy that is different in honey bees compared to most other insects is the ovipositor, or the egg-laying organ. In most other insects, the ovipositor is used to lay eggs; in honey bees, however, it has adapted to become a stinger. This is why only female bees can sting: Their stinger is a modified egg-laying organ! Instead of laying eggs from the ovipositor, honey bees lay eggs from the vagina.

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Quiz: Honey Bee Reproduction

You must achieve a score of 100% on this quiz to complete the course. You may take the quiz as many times as you need to achieve that score.

Please complete this activity in the course.

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Discussion: Module 2 Discussion and Questions

This course provides the opportunity to discuss content with fellow beekeepers in addition to the course instructor. You are welcome to participate in all course discussions and are **required to participate in this discussion to pass the course**. Meaningful participation includes the following options:

- Providing thorough answers to the questions posed by the instructor
- Helping answer other students' questions
- Sharing additional information on the topic based on your prior experiences or knowledge

Please answer the following questions about **Module 2: Honey Bee Reproduction**:

- The growth and productivity of a colony is directly influenced by the queen. In recent years, there has been a rise in beekeeper concern about queen quality. One measure of queen quality is the number of unrelated drones with which she has mated. Based on what you've learned about mating and reproductive systems, what are some of the things you could do to improve queen quality in your beekeeper community?
- What questions do you have for the instructor related to the information provided in this module?

Instructions:

- Click **Reply** to post a comment or reply to another comment.
- Please respond in a text format or as a video using the filmstrip icon that is available once you click **Reply**. (If posting a video response, we recommend that you do not use your cell phone as most do not use Flash software, which is required to convert the recording.)

Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [Cornell's policy regarding plagiarism](#) (the presentation of someone else's work as your own without source credit).

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Module Wrap-up: Honey Bee Reproduction

In this module, you examined the dual nature of honey bee reproduction, exploring in detail how individual bees mate and produce offspring within the colony. Understanding these fundamental biological processes and behaviors is essential for successful colony management and provides crucial insight into the complex reproductive dynamics of honey bee societies.

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Module Introduction: **Honey Bee Development**



All bees begin life as an egg. From there, some become workers, others queens, and others drones. As adults, bees continue to change in response to the social and physical environment of the colony and the natural world. In this module, you will learn all about how bees develop and how their physiology and behavior change throughout their lives.

Photo credit: Lucinda VanVleck

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Watch: The Stages of Development

Honey bees go through an ordered pathway of development as they transition from egg, to larva, to pupa, to adult. Although there is slight variation in development time from bee to bee, one is able to generally track how many days it takes queens, workers, and drones to mature through the various phases. Being able to identify the different stages of development, knowing how long each stage lasts, and predicting the emergence of the different castes will come in handy for you as a beekeeper. Such knowledge is directly applicable, for instance, to queen rearing and to management practices that reduce *Varroa* mite loads.

Watch this video for more about how honey bees develop from egg to adult.

Video Transcript

So by now, as a beekeeper, you're probably familiar with the basic development of honey bees. Honey bees begin life as an egg that hatches into a larva that becomes a pupa, and then undergoes metamorphosis to form an adult. But it's really important, as a master beekeeper, to know how the developmental timing differs among the castes. So for all three castes, they all take three days as an egg and six days as larva. The amount of time as a pupa is what is different. Drones take 15 days to pupate, workers takes 12 and queens only take seven days despite the quality and quantity of resources that goes into creating them.

So why would this be? There's actually a few reasons why natural selection would favor a quick development time in queens. So think about a time when a colony is producing lots of queen cells, like when they're swarming. The first queen to emerge is going to be the queen for that colony, and as soon as she emerges she's going to kill all the other rival queens. So it's actually advantageous to develop as quickly as possible. Also developing quick as a queen gives you a little bit of a headstart to start growing your colony, collecting food and gets you better prepared to survive winter. And even though this might be a difference of a day or two, this can make an impact if it's perfectly timed with a nectar flow. And lastly, if a colony finds itself suddenly queenless, if the queen dies or something else happens to her, then they need to produce a new queen quickly. And if you have a quick development time for queens, this ensures that the colony isn't queenless for very long.



Okay, so what is exactly happening during development? Three days after the egg is laid it hatches and a small larva crawls out. And a one-day-old larva is actually really tiny. It's about the same size as an egg, but you can detect it with your eye. The main function of a larva is to eat. In fact, its stomach is almost the entire length of its body. And it's fed many meals a day by nurse bees. Workers are first fed royal jelly and then they're fed bee bread, which is pollen mixed with honey and some saliva and fermented by microorganisms. After it's eating, it's growing, and it needs to shed its skin to accommodate its bigger body. So after six days of eating, growing, molting, eating, growing and molting, it's finally ready to pupate. And during this time, workers are going to cap the cell with wax, the larva spins a cocoon around its body, and it doesn't eat anymore until it becomes an adult.

While it's a pupa, it starts to develop features that make it look more like a honey bee and less like a grub. Its three body segments — the head, thorax, and abdomen, are becoming distinct. And you can start to see its mouth parts, its eyes, its legs and antennae. And the eyes will start to take on color. Instead of being white they start to turn purple, which is referred to as the purple-eyed stage of development, and then they'll eventually become black. And after seven to 15 days, depending on if it's going to be a queen, a worker or a drone, it has its final molt and it chews its way out of its cell. It actually leaves its cocoon behind in the cell, and it's ready to begin life as an adult bee. And you can actually identify these one-day-old bees that have freshly emerged in the colony, because they have really fuzzy heads and thoraces, and they're actually not able to sting at this stage. So they're pretty cute.

To witness and understand the full process of bee development (including under the pupal cappings!), [**watch this video by Anand Varma**](#). It shows the 21 days of a worker's development from egg to emergence, condensed into one minute.

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Tool: The Stages of Development Timeline

Take a look at the developmental timeline tool to get a better sense of the stages of development for all honey bees.



Download the Tool

Use **this chart** as a developmental timeline reference for honey bees.

Understanding developmental timing can give you information about the queen's fertility, which is more valuable than actually seeing her in the hive. The presence of queen-laid eggs is a sign that the colony's queen has been there in the past one to three days, so she is most likely in the hive. If there are no eggs or young larvae during a time when you would expect the queen to be laying, but you see older larvae, you can infer that it has been six to nine days since she has laid any eggs. You should then do a thorough check to see whether she is in the colony and anything else you can learn.

Similarly, understanding the timing for queen emergence is critical if you are selling queen cells or introducing them into queenless colonies. You want to make sure that emergence is timed for a day or two after they have been introduced to a colony, rather than before!

You'll notice in this course that we simplify the amount of time it takes the different colony members to pupate. Scientific observations find queens take an average of five to six days, workers take six days, and drones take six to seven days. In this tool, we report all colony members take approximately six days to pupate, but just know there can be some variation.

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Read: How Sex Is Determined

What determines whether a bee develops into a male or a female?

Honey bees, as well as the other insects in the order Hymenoptera, have a different sex determination system than humans.

In humans, both males and females inherit two sets of each of their chromosomes in their cells, one set from each parent. One particular pair of chromosomes is the sex

chromosomes, and these are what determine a person's sex. A person inheriting an X chromosome from its mother and a Y chromosome from its father develops into a boy, while a person inheriting an X chromosome from both its mother and its father develops into a girl. In honey bees, chromosomes also play a role, but there are interesting differences.

Before we describe the role chromosomes play, let's talk about the ratio of male to female brood produced in a colony.

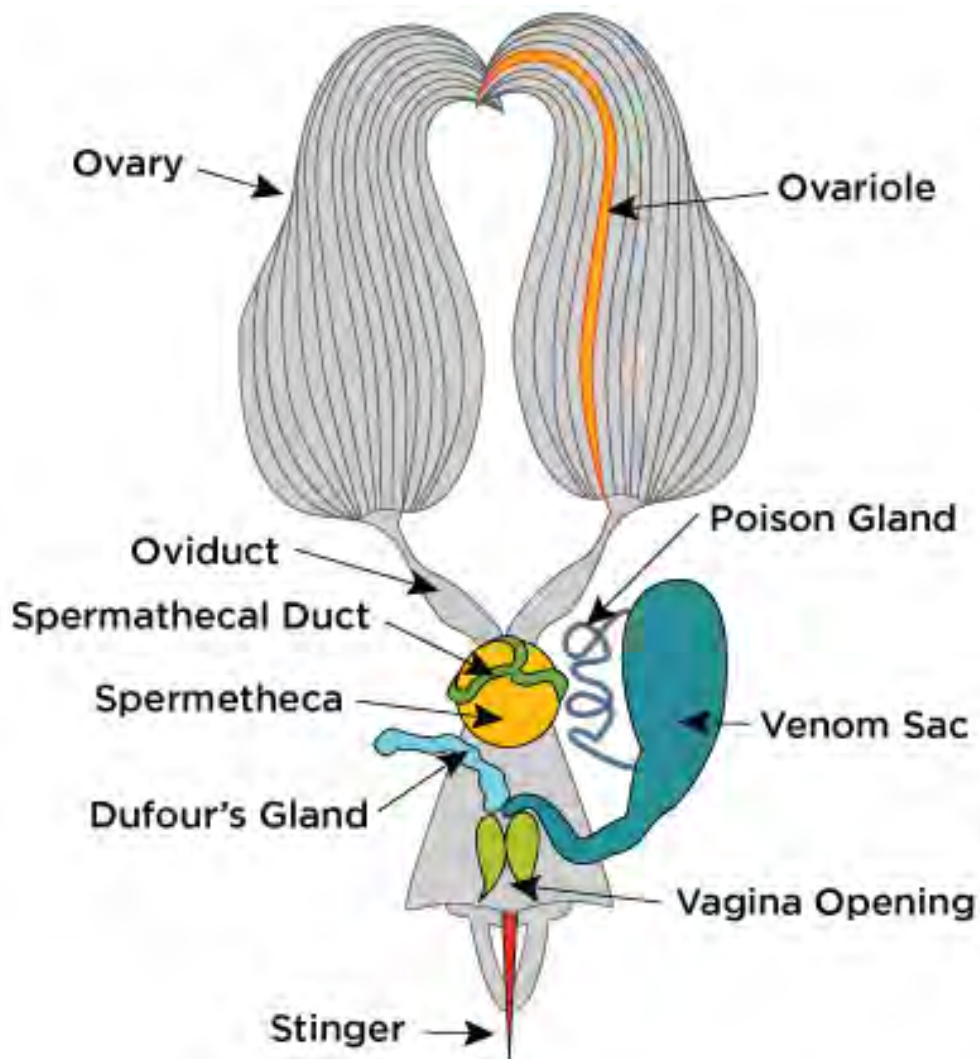
The queen has control over which eggs she fertilizes by opening her spermathecal duct to release sperm or keeping it closed so no sperm is released. So what makes her fertilize an egg or not? It is actually the worker bees that build comb that affect the ratio of worker offspring to drone offspring in the colony. They build large cells (approximately 6.4 mm wide) to accommodate drones and small cells (approximately 5.2 mm wide) to accommodate workers.

☆ Key Points

The ratio of females (workers) to males (drones) in the colony is dynamic, depending on colony size, comb architecture, and queen fertility.

Inbreeding can result in high numbers of diploid males because of the sex determination system in honey bees.







Drone cells (bottom) are slightly larger in diameter than worker cells (top)

The ratio of workers to drones depends largely on the population of the colony. Once the colony has at least 4,000 or so workers, it can begin investing in drone production, and comb builders will begin constructing larger cells for drones. The queen walks along the combs and assesses which type of egg she should lay in each cell based on its size. She sticks her head and thorax into empty cells and measures their size using her front legs. Small cells stimulate her to lay a fertilized egg, whereas large cells stimulate her to lay an unfertilized egg.

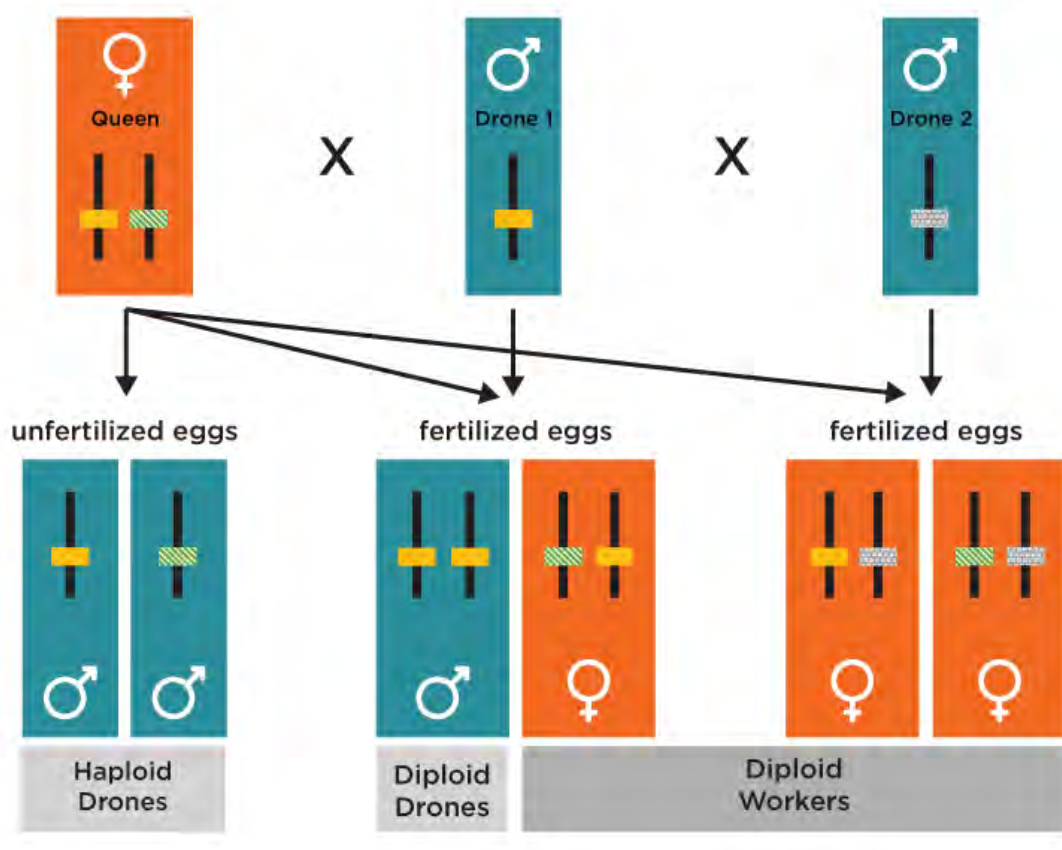
Honey bee sex determination is usually explained simply in this way: Fertilized eggs yield offspring that inherit two sets of chromosomes (so are diploid organisms) and develop into females, while unfertilized eggs yield offspring that inherit only one set of chromosomes (so are haploid organisms) and develop into males. However, it actually goes one step beyond this. Let's explore the Complementary Sex Determination gene next.

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Read: The Complementary Sex Determination Gene

It is not the number of sets of chromosomes a bee inherits that determines their sex; rather, it has to do with the gene version (also called an allele) they possess at a specific gene position on a certain chromosome. The special gene is the **Complementary Sex Determination (CSD) gene**, and this is what ultimately determines sex in honey bees. If a bee receives just one copy of this gene (from the mother's egg), then a male is created. Yet if a bee receives two copies of this gene (one copy from their mother's egg and one copy from their father's sperm), then it generally means a female is created. Here is the key, however: *A female's two copies must be different versions.* If a bee receives two different versions of the gene, then she develops normally as a female, but if the two versions happen to be identical, the bee will develop as a male — even though it has two sets of chromosomes. We call these "diploid males," but you rarely see diploid male adults walking around the colony because they usually get detected and cannibalized by their sisters while they are still larvae.



Sex determination in honey bees depends on the CSD gene (horizontal yellow, green, and gray rectangles) located on a specific chromosome. A colony formed from a queen that mated with two drones yields six potential offspring. The number and version of the gene inherited determines each offspring's sex.

There are several implications this has for the colony. If many larvae receive the same two versions of this gene and get removed, the brood pattern will end up looking spotty. Populations with largely unrelated honey bees usually have higher genetic diversity. In these instances, there are many versions of the CSD gene in the population, and the chances of a bee inheriting two identical versions are low, so diploid males are rare. Yet in populations that are heavily inbred, with low genetic diversity, the likelihood of inheriting the same version of the CSD gene is much higher. These highly inbred bees will likely produce a spotty brood pattern because the workers are removing so many diploid males. For this reason, it is important that honey bee breeding programs do not result in inbreeding.



Spotty brood pattern can be the result of inbreeding. Photo credit: Bee Informed Partnership

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Read: Diploid Male Occurrences

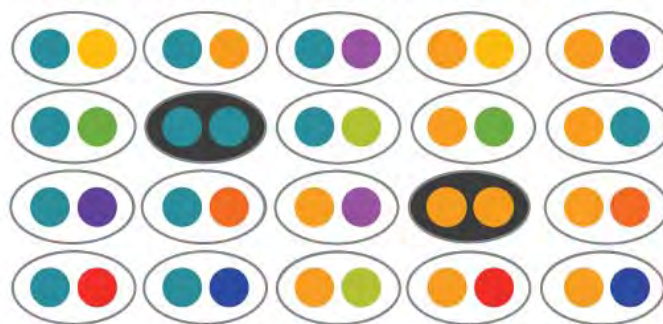
If spotty brood can indicate health issues in your hive, how many larvae can we normally expect will receive two copies of the same gene in a normal, healthy population?

The number of different versions of the CSD gene varies in normal populations, but studies have found it tends to range between six and 17.

Alleles in the population: 10 (range from 6-17)



If she mated with males that all had different alleles, 2/20 (10%) of her potential offspring would die.



Let's imagine one population has 10 different versions of the CSD gene. Each of the colored circles in the above illustration represents different CSD gene versions.

Our queen has two of these 10 possible versions, one on a specific chromosome she inherited from her mother and the other from her father. So let's say our queen mates with 10 males, and each male has a unique version of this gene.

We can draw all the potential fertilized offspring this queen can have. In this scenario, there are 20 different combinations that are possible, and of all these combinations, there are two cases where the offspring inherits identical gene versions. These two offspring — representing 10% of all the offspring the queen lays on average — would be the ones that are removed and cannibalized. This goes to show that in normal populations, we can always expect to see some spotty brood due to the sex determination system in honey bees. It would be rare to find an absolute perfect brood pattern on a frame.

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Watch: How Caste Is Determined

For generations, people have wondered what determines whether a bee will become a queen or a worker. How is the decision made? What factors are involved? There are still some mysteries surrounding the exact molecular changes that occur during the process of caste determination, but scientists are getting close to piecing together the whole picture. This video addresses our current understanding of honey bee caste determination.

Video Transcript

Interestingly, caste determination in honey bees is due to environmental factors instead of genetic factors. This means that a young, female larva can become either a worker or a queen. Think about that for a moment. They differ vastly in their appearances and in their behavioral repertoires and an order of magnitude of a difference between their lifespan. Workers only live for a few weeks and queens live for a few years. Yet they arise from an identical DNA blueprint. The key is that there is a short time period during which a larva determines to either become a queen or a worker. And this time period is between 12 and 36 hours, from when it hatches as an egg. So anything beyond 36 hours the larva has already been destined to develop the way that it was meant to whether it's a worker or a queen, there is no switching.

And this has implications for a colony if it suddenly finds itself queenless. If the queen dies, workers need to try and rear emergency queens out of larvae that is already existing in the colony. And they could only do this from larvae that are less than 36 hours old. So if there aren't any young larvae in the colony it's considered hopelessly queenless unless a beekeeper intervenes and introduces a new one or a frame with young brood on it. So what determines this developmental pathway of a larva? What is that environmental factor? It's actually royal jelly. And royal jelly is a food that's secreted from the hypopharyngeal glands and the mandibular glands of nurse bees. It's milky in its consistency and it contains water, sugars, salts, fatty acids, other lipids, vitamins, free amino acids, and proteins. So basically it's highly nutritious.

And queens and workers actually receive different amounts of royal jelly, and this contributes to why they are different castes. So queens receive royal jelly for their entire period as a larva, all six days. But workers are fed royal jelly only for the first two to three



days as a larva, and then they transition to eating bee bread. Also, queens are fed approximately ten times more often than worker larvae are fed. About 1,500 times, versus 150 times. So, essentially, some components of royal jelly are starting a cascade of reactions in the bee's cells that begin to drive this differential development. And it turns out that there are hundreds of genes that are involved. Hundreds of genes are being turned off and turned on, and worker larvae, and queen larvae as they're developing. And although queens and workers have the same DNA blueprint to begin with, the take-home message is that genes of this blueprint are turned on and off differently. And these can be genes that are involved in life span or body size or other factors as well, like reproduction, and there's just many that are involved and it's a very cool and complex process.

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Read: Molecules and Food Involved in Caste Determination

There are slight differences in the jelly fed to workers and queens. The popular name for all milky brood food, whether it's fed to queen or worker larvae, is "royal jelly." It's worth noting, however, that scientists have determined that, while the substances are largely the same, there are slight differences between the jelly fed to workers and the jelly fed to queens. You may sometimes hear the terms "worker jelly" and "royal jelly" used to differentiate between the two.

Royal jelly has a lower moisture content and a higher sugar content than worker jelly. It also contains higher amounts of a fatty acid called 0-hydroxy-2-decenoic acid. The jelly fed to different-aged brood changes slightly over time as well; e.g., the moisture content, sugar content, and mineral content in the jelly are different depending on whether it's fed to a larva that is one day old, two days old, three days old, and so forth.

The consumption of royal jelly triggers responses in hundreds of genes and proteins; many hormones are also involved. Caste determination is complex and dynamic, with scientific research still unfolding. For the purposes of this course, we will cover only two components that are well studied for their role in caste determination.

Juvenile hormone is a hormone found in all insects, but it plays an important role in honey bee caste determination. The hormone is high in emerging queens and low in emerging workers. Eating lots of royal jelly as a larva stimulates a high production of juvenile hormone, which in turn contributes to the development of a queen with large ovaries comprised of many ovarioles, as well as a long lifespan.

Vitellogenin is a protein that is produced by the fat body in the honey bee and found in the ovaries, among other tissues. In the ovaries, it functions to help make eggs. Larvae with high vitellogenin levels in their hemolymph develop into queens. Consuming royal jelly as a larva increases the level of vitellogenin in the developing queen, ensuring that she can create many eggs throughout her lifetime. Vitellogenin levels in queens remain high as adults, which likely plays a role in contributing to their long lifespans. On the other hand, larvae with low vitellogenin levels in their hemolymph develop into workers. As adults, this protein remains at low levels in their ovaries, contributing to their reduced fertility.





The quantity of royal jelly a larva consumes initiates cellular reactions that lead to the development of either a queen or a worker bee.

Photo Credit: Eleanor Andrews

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Read: Inter castes

A queen is successfully reared from a female larva under three conditions:

- When the larva is **exclusively** fed royal jelly
- When it is fed **copious** amounts of royal jelly
- When the larva is selected at a **young age** to start receiving this special queen diet

If even *one* of these three conditions isn't met, the resulting queen that develops is an "intercaste." Inter castes can develop when they are fed small amounts of royal jelly as larvae, when their diet is altered to differ in the ratio of sugars or other nutrients that are naturally occurring in royal jelly, or when the larva that starts receiving a queen diet is older. Inter castes can develop from larvae that are as old as four days!

Inter castes have characteristics of both queens and workers. The length of their abdomen, the size of their spermatheca, and the number of ovarioles for inter castes are between those of a worker and a queen. Some inter castes develop corbicula on their legs for pollen storage, while others develop with a shriveled spermatheca and are therefore incapable of storing sperm. If an intercaste develops with more queen-like features than worker-like features, they can act as a queen for at least a short period of time in the hive — mating, laying eggs, and attracting a retinue of workers — but they tend to be superseded by workers in the same season.

Naturally occurring inter castes are rare in a colony and only result if workers are forced to rear an emergency queen from older larvae. They can occur, however, if beekeepers are grafting older larvae when they are queen rearing.

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Watch: How the Worker's Role Transitions Throughout Its Life

The tasks that workers specialize in allow the colony to perform many functions at once. The nest can be constructed and cleaned, food can be collected and stored, brood can be fed, and the colony can be guarded - all at the same time! This is made possible because workers specialize in different tasks at different times. As a worker ages, she learns and performs different roles. Watch this video to learn how tasks change over a worker's life.

Video Transcript

The queen and worker castes represent a division of labor and reproduction. But this isn't the only division of labor we see in a colony. Labor is also divided up among workers who transition through many tasks or duties as they age. And this transitioning and specializing in different tasks at a time is called polyethism. And the tasks that they perform depend on the age of the bee and also the needs of the colony. So roughly speaking, workers begin their first half of their life as house bees, performing tasks inside the nest and then they spend their second half of their lives as field bees, performing tasks outside the nest.

The very first thing that a bee does when she emerges from her cell is she cleans her cell and cleans the other cells around her. Then she transitions into capping brood and tending to the brood where she's called a nurse bee or attending to the queen where she's called the retinue, or makes up the retinue. Then she transitions into receiving nectar that's coming into the colony, cleaning debris in the colony, packing pollen and comb building. So just general hive maintenance. When she's finally ready to start transitioning outside of the hive, they will perform tasks that are right in the intermediate, right at the hive entrance. So they can participate in ventilation at the entrance or guarding the entrance against intruders. And then when they're finally ready they take their first foraging flight. And as foragers they can collect a variety of different things. They can forage for nectar, pollen, water, resin or even new nest sites if a colony swarms. And they usually remain as foragers until they die of natural causes.

A really neat thing is that foragers specialize in what they're collecting. So they may be specialize as water collectors or as propolis collectors, although they can be flexible if the need arises. It's really interesting to note that there is a variation as to when a worker is beginning a new task. So, a worker that's finally becoming a forager might do that after 14



days, whereas another worker might do that after 19 days. It's not so rigid or set in stone. It's flexible. And the really interesting thing to know as a master beekeeper, and where it's relevant, is that workers with different duties are found in different parts of the nest. So nurse bees are usually found on the brood comb, because they're tending to the brood. And that's usually toward the bottom or center of the hive. Guards are usually found at the entrance of the hive. And comb builders are found wherever there's new comb being built, which could basically be anywhere within the colony. And when home, foragers can also be found throughout the colony. but they're most often clustered below the brood area, so that they can easily enter and exit the hive. And this is important to know because it might impact your decisions when you're inspecting colonies, when you're monitoring for certain parasites or diseases and also for swarm control.

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Tool: Worker Bee Ages and Tasks

As you just learned, the worker bee's role transitions throughout its life. This tool will help you understand the average ages that workers perform each task.

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Download the Tool

Download this tool for more information on polyethism.



Read: Molecules Involved in Worker Polyethism

The structures and chemicals in honey bee brains change as they age and as they adopt new behavioral repertoires. The shifts in worker duties are regulated by hormones, proteins, and neurotransmitters within the bee's hemolymph and neurons. Some of these molecules may be familiar to you, as they are also involved in caste determination. It is not uncommon in biology for molecules to be involved in more than one function.

Changes in worker duties are caused by subtle changes in the levels of these and other hormones and proteins, which are in turn regulated by the social environment of the colony. The presence and fertility of the queen, the population level, the amount of brood present, and the amount of food coming in, can all influence what task a worker is going to perform.

Here we will touch upon four molecules involved in worker polyethism:

Juvenile Hormone

JH (Juvenile hormone) influence the transition from in-hive tasks (like nursing) to outside tasks (like foraging). Newly emerged worker bees have low levels of this hormone in their bodies, but the level increases steadily throughout the first 15 days or so of a worker's life. At this point, juvenile hormone assists in the shift from tasks inside the hive to tasks outside, thus promoting the shift to foraging.

You may be wondering why this hormone is called 'juvenile' hormone if it is actually highest in bees. It turns out, the progression of juvenile hormone levels are actually opposite in honey bees compared to other insects. In other insects, this hormone is highest in juveniles and decreases as the insect ages into adulthood.

Vitellogenin

In addition to facilitating egg production in the ovaries, vitellogenin is present in the hypopharyngeal glands, the fat body, and the hemolymph. In the hypopharyngeal



glands of nurses, vitellogenin helps create royal jelly to feed to brood. Nurse bees have high levels of vitellogenin in their hemolymph and their hypopharyngeal glands, and the levels decrease as bees age and become foragers. This makes sense because it is nurses that feed larvae, while foragers do not tend to the brood. It is this rapid decrease in vitellogenin, combined with the increase in juvenile hormone, that is associated with the behavioral switch from nurse to forager.

Dopamine

Dopamine is a neurotransmitter that is present in many organisms, including humans. Neurotransmitters are chemicals released by neurons in the brain to send fast signals to other neurons. In honey bees, low levels of dopamine stimulate nurses to care for brood. So which bees in the hive likely have low levels of dopamine? You guessed it – younger nurse bees. Their levels are much lower than those found in foragers. It is the presence of the queen (and her pheromones) that regulates dopamine production in worker bees' brains. Nurse bees (rather than foragers) come in contact with queen pheromone, which sets off a molecular pathway in the nurse bee brain to reduce the production of dopamine and thereby stimulate nursing behavior.

Pheromones

Several pheromones are known to play a role in the regulation of workers' role transitions throughout their life. For example, the foraging pheromone and especially one of its components, ethyl oleate, is emitted by returning foragers and it suppresses a premature switch to foraging roles in young bees. Here is another example: the brood pheromone emitted by the developing larvae signals the need for nursing. This pheromone can delay the onset of foraging behavior in worker bees, ensuring that the brood is well-cared for by young bees. These, and other



pheromones, assist workers in their assessment of the needs of the colonies (more foragers? more nurses? etc.) and they contribute to the remarkable flexibility exhibited by honey bees when it comes to role allocation in the colony.

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Read: Social Environment Influence on Worker Tasks

How does the social environment influence worker tasks? Although the molecules involved in worker polyethism change as a bee ages, and this influences the transition to different tasks, age is not the only factor that determines when workers will be house bees and when they will be field bees. If that were the case, colonies wouldn't be able to respond to changes in their environment. So, in addition to age, there are also strong effects of the social environment.

Honey bees are able to detect what is going on inside the nest and alter their behavior accordingly. For example, although house bees transition to become field bees at around 15 days of age, the exact age at which this happens is adjustable. If a high percentage of the foragers are removed from a colony (e.g., through a pesticide poisoning event), their sisters will recognize that the colony no longer has enough bees to bring in pollen and nectar. Some nurse bees will transition to becoming foragers earlier than they would otherwise. In the reverse, if the nurse bee population suddenly declines, some foragers will revert to being nurses.

More broadly, worker bee behavior is flexible enough to take on tasks that are needed to help the colony run smoothly. Worker bees spend a considerable amount of time performing tasks, but they actually spend most of their time being inactive and walking around the nest. It is likely that they are assessing their colony's state at this time, sensing social and environmental cues of the colony condition. This ability to read the colony's situation and then respond accordingly makes these honey bee societies flexible and resilient. For example, during a nectar flow, foragers will recruit more bees to help receive the food through a tremble dance, and brood rearing increases. When there is not enough storage space for holding the incoming nectar, more bees will engage in comb building.

A remarkable example of this ability to detect and respond to a colony's needs is the work of heater bees. Heater bees are workers found in the brood nest that generate heat through shivering as a way to incubate pupae at the proper temperature. While brood needs to develop at a certain temperature (91°F to 97°F) to become healthy adults, they can tolerate slight variations. Interestingly, these small variations can impact the behaviors of these bees after they emerge. For instance, bees that develop at the highest end of the range (closer to 97°F) transition to become foragers at an earlier age, are more likely to



perform undertaking tasks, are better learners, and perform more dances than bees that develop at the lowest end of the range (closer to 91°F). Beyond this range, however, pupae fail to develop properly or die.

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Read: Bee Lifespan

While healthy queen bees live on average two to three years (a very long time for an insect!), worker lifespans are a lot shorter. Their lifespan depends on a number of factors, including their genetics, the activities performed throughout their lives, and the season. During spring and fall, workers typically live for one to two months. In the summer, they live much shorter lives, typically two to five weeks. Bees that live throughout the winter are the longest living bees. These bees are called winter bees, and can live for four to five months! They are needed to survive over winter so they can regulate the temperature in the brood nest and help rear the first brood at the end of winter.

Winter bees are generated within the colony in autumn from the end of August through November and they differ in their physiology compared to summer bees. Winter bees have lower levels of juvenile hormone (analogous to nurse bees who spend their time performing in-hive tasks) than summer bees, and they have much larger hypopharyngeal glands and larger and more numerous fat body cells than summer bees. The fat body is a collection of tissues comprised of many fat body cells that store fat and glycogen (stored sugar) to release for energy production in times of food shortage. It is located mostly in the abdomen, but also spans the thorax and head.



Drones that do not mate with queens will live an average of three to four weeks. Their lifespans may be shortened as winter approaches because workers usually evict drones from the hive prior to winter. The colony does not expend precious food stores on feeding drones that do not aid in survival over the winter

Drones usually get evicted by their sisters in the fall.

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Quiz: Honey Bee Development

You must achieve a score of 100% on this quiz to complete the course. You may take the quiz as many times as you need to achieve that score.

Please complete this quiz in the course.

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Discussion: Module 3 Discussion and Questions

This course provides the opportunity to discuss content with fellow beekeepers in addition to the course instructor. You are welcome to participate in all course discussions but are only *required* to participate in two specific discussions (Module 2 and Module 6.)

This Module 3 discussion is optional and your participation in it does not count toward your final grade. We encourage you to read other students' comments and to respond if you are interested.

In this discussion, you'll reflect on honey bee development timing and its management implications.

1. Write an initial post in which you share and analyze bee development, using the following questions to guide you:
 - How do development timelines affect your colony management decisions?
 - What specific management practices will you adjust based on this knowledge?
 - What environmental factors influence your development timing calculations?
 - What challenges do you anticipate in timing management activities?
 - How might regional or seasonal variations affect your approach?
2. Aim to return to this discussion, read your peers' replies and respond to them. Consider the following questions:
 - How do management approaches differ across climate zones?
 - What common challenges do beekeepers face with development timing?
 - What patterns emerge in how different operations handle timing issues?
 - What practical tips from other beekeepers might improve your management?

To participate in this discussion:

Click **Reply** to post a comment or reply to another comment. Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [eCornell's policy regarding plagiarism](#) (the presentation of someone else's work as your own without source credit).

Please note:

While discussion board postings will be accepted through the end of the course, we strongly encourage everyone to move through this course as a group. As such, postings



that are made to this board after 5pm ET on the due date will be read but may not receive a response on the board from your facilitator and/or your peers. Please let your facilitator know if you have any questions.

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Module Wrap-up: Honey Bee Development

In this module, you examined the remarkable journey of honey bee development, from egg to adult. You learned how environmental and social factors determine whether eggs develop into workers, queens, or drones, and how adult bees continue to adapt and change their roles throughout their lives. This knowledge of bee development and physiology reveals the dynamic nature of colony life and the impressive adaptability of honey bee behavior.

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Module Introduction: **How Honey Bees Communicate**



If you think about it, you'll realize that all living things need to communicate. Birds sing, wolves howl, and even fish use body language. We typically think communication is all about our own uniquely complicated use of the voice, but most forms of life use other sensory modalities to communicate. It is really no surprise that eusocial insects like honey bees are elaborate, multi-sensory communicators. Even though we only understand a fraction of their language, that tiny part is fascinating!

Bees live together in the dark. It is reasonable to assume that vision is not a sense they rely heavily on for communication. Instead, bees mostly use the senses of “hearing” (vibration detection), taste, smell (pheromones), and touch to “speak” with each other.

The famous biologist Karl von Frisch first described a type of bee communication that he called a “dance” in the 1940s, work for which he won a Nobel Prize. Subsequent studies of different forms of physical bee communication — all called “dances” — were conducted by a number of his students and others, such as Martin Lindauer, Jürgen Tautz, and Tom Seeley. In 1960, the communication signal “Queen Mandibular Pheromone” was discovered, opening the doorway to the captivating world of bee pheromones. In the years since this discovery, over 28 uniquely different honey bee pheromones have been described.

In this module, you will learn some of the ways honey bees communicate important information through pheromones and dances, and you will learn how to recognize different communication behaviors in your own colonies.

Photo Credit: Lucinda VanVleck

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Read: Pheromone Communication

Honey bees have evolved one of the most complex systems of pheromone communication known in nature.

This communication regulates functions ranging from reproduction to foraging, enables communication among the colony members within a hive, and occasionally permits communication between colonies. The pheromones emitted by an individual

bee can change over the course of its lifetime or as a function of its reproductive status. Of the many chemical signals that bees are known to emit, only a small fraction of them are well understood.

☆ Key Points

The honey bee colony is a chemically busy environment, with pheromones secreted by the queen, brood, and workers.

Pheromones elicit behavioral changes in members of the same species.

▼ Queen-Produced Pheromones

Let us start with the most important pheromones, the ones produced by the queen. Why are those the most important? First, it is known that when the queen is old or sick (low pheromonal signal) or it dies (no pheromonal signal), workers are driven to rear new queens from young brood within 12–24 hours. Second, the removal of the queen in absence of young brood soon leads to the decline of the colony: the workers stop performing their activities and start to lay unfertilized eggs that develop into male adults (drones); the colony becomes disorganized, unfit, dirty, susceptible to diseases and prey of predators; it rapidly depopulates and goes toward a certain death.

The **queen signal** includes 40-100 compounds produced from multiple glands: the mandibular gland, tergal glands, Dufour's gland, Koshevnikov gland and tarsal glands. Those chemicals attract drones during mating flights. They stimulate foraging and brood rearing, while inhibiting queen rearing and



suppressing ovary development in workers. They stabilize swarms (along with the Nasonov pheromone produced by workers), and much more. How can ONE individual affect the physiology and behavior of thousand of individuals dispersed in the hive? The answer is that the queen signal attracts workers to surround and tend to the queen. This retinue of workers that are constantly touching and licking her pass along the pheromone to the rest of the colony through body contact, antennating (touching antennae), and trophallaxis. The transfer of these pheromones enables the bees in the hive to know that the queen is alive and fertile, thus providing social cohesion to the colony.



Workers detect queen pheromone by licking and antennating the queen. Photo credit: Bee Informed Partnership.

Five of the dozen components of the queen signal are secreted by the mandibular gland; these are collectively known as **queen mandibular pheromone** (QMP). This is the best-understood, behaviorally and chemically, pheromone in honey bees. A strong signal indicates that the queen is healthy and productive, which inhibits the development of workers' ovaries. It also attracts workers to surround her in the hive or in a swarm and stimulates them to tend to her. Queen mandibular

pheromone also helps regulate the transition for nurse bees becoming forager bees by delaying the onset of foraging.

What about the other compounds emitted from other glands? They are less well known, behaviorally and chemically. We know about the queen's **Dufour's gland** which is located near the egg laying device of the queen, aka the ovipositor. As the queen lays an egg, she secretes a unique waxy, non-volatile, mix of ~ 20 compounds that ends up covering her eggs. This unique chemical signature allows nurse worker bees who inspect the eggs later using antennal contact/mouthpart contact to identify queen-laid eggs. Why is that important? Because this allows nurse workers to identify worker-laid eggs, since those do not have the same chemical signature. As you know, workers are not supposed to lay eggs, and there are many pheromones present in the



hive to deter them from having functional ovaries, yet every now and then, and more under some situations, workers lay a few eggs. Those eggs do not bear the queen's chemical signature thus workers destroy them. This is called worker policing.

Another portion of the queen signal that has been characterized is what oozes from the queen's tarsal glands in the last segment of her legs. The **footprint queen's** secretion is oily, colorless and composed of ~ 12 non-volatile substances that are extruded through openings on the tarsi when the queen is walking onto the comb, as she explores the hive for empty cells she can use to lay her eggs. Those secretions deposited as the queen walks throughout the hive signal to the colony that there is a functional queen present and that she is active. This in turn maintains colony stability, contributes to the inhibition of queen cup construction by workers and also to the inhibition of worker reproduction.

▼ Brood-Produced Pheromones

While larvae do not seem to do much other than eat and grow, they actually influence the behavior of nurses and help ensure their own care through **brood pheromone**. This pheromone is released by the larvae and detected by the nurse bees. It gives the nurses information about a specific larva's sex, caste, and age, which in turn influences the nurses to provision them different amounts and types of food. It also inhibits worker ovary development and can affect the onset of foraging in nurse bees. Young nurse bees that detect brood pheromone delay their transition time to becoming foragers, while older nurse bees that detect the pheromone are stimulated to "speed up" their transition to foraging.



Larvae release pheromones to provide nurses with information.



▼ Worker-Produced Pheromones

The **Nasonov pheromone** is the primary means of aggregation and orientation for honey bees. The pheromone is released from the Nasonov gland, located near the end of the abdomen in workers. There are three primary behavioral contexts where the Nasonov pheromone is used: marking the nest entry to assist returning bees, aggregation during swarming, and marking a water source. The ability of adult workers to release this pheromone increases with age; peak production occurs when the bee is about 28 days old. This pheromone is unique to workers, as queens and drones do not contain a Nasonov gland.

Synthetic Nasonov pheromone has been manufactured into a lure that can be used by beekeepers to attract swarming bees to unoccupied hives. Some of the same compounds in Nasonov pheromone are also found in lemongrass oil, so beekeepers can also use lemongrass oil to increase the chances of catching a swarm.



Worker bee exposing her Nasonov gland.

Photo credit: Bee Informed Partnership

The **forager pheromone** is derived from fermented nectar, which foragers regularly consume when they are out collecting resources. They process a chemical in the nectar and exude it through their exoskeleton. Bees inside the hive smell this chemical when the foragers return. This pheromone delays the onset of foraging in nurse bees: When there are numerous foragers and therefore high levels of the pheromone, the transition is delayed. By contrast, when there are few foragers, the level of this pheromone is low, and some nurse bees are stimulated to transition into foragers to help maintain the field force.



▼ Alarm Pheromone

When a worker bee is disturbed by a predator or threat to the colony, she emits **alarm pheromone** to mark the location of the potential threat and to rally others to aid in defense. She releases alarm pheromone from her mandibular gland and a gland beside the stinger. When stinging, the gland near the stinger produces the pheromone and it coats the sting shaft. When the bee stings the enemy, the alarm pheromone "marks" the sting position so other bees can quickly locate it. Promptly removing the stinger and smoking the area helps to mask the smell of this pheromone to prevent future attacks. Note that dark colors, erratic movements, and certain odors, such as leather, can elicit an attack response prior to alarm pheromone being released.

Alarm pheromone is one of the honey bee pheromones that is easily smelled by humans. If you've smelled it before, you may have thought it smells a bit like bananas. Interestingly, one of the main components in alarm pheromone, isoamyl acetate, is the same compound that gives bananas their sweet smell.



Alarm pheromone coating the sting shaft helps mark intruders for easy identification by other bees.

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Watch: Honey Bee Dances — The Waggle Dance

The various forms of physical (touch) communication among honey bees are called “dances.”

Karl von Frisch described what has become known as the “waggle dance,” a behavior used by a successful forager to share information about the location and scent of a rich food source with other foragers.

In this video, Dr. Tom Seeley, the Cornell professor who has dedicated much of his career to studying honey bee communication and behavior, describes this fascinating behavior.

Video Transcript

Okay, hello. What I want to share with you now is the waggle dance of the honey bee. This is a very special communication signal. And what makes it special is that it's a signal that honey bees produce that shows other individuals the location of a rich food source and there's only one other species that has that ability with the honey bee. That's ourselves, human beings. We do that when we give somebody directions to find a good grocery store, for example. And the honey bees are doing that to show their hivemates the locations of rich food sources, flower patches.

Now, one thing I'd find helpful in explaining the waggle dance is to understand that you can think of this as the bee performing a miniaturized reenactment of her flight out to the flower patch. And when I say a miniaturized reenactment of the flight, that's actually realistic because as the bee goes through the movement of the waggle dance which consists of her doing the waggle motion in the center of the formation of the dance and then she circles back and does another waggle run. She's also buzzing her wings. And she is pointing you in the direction that indicates the direction to the food source and the length of that waggle run, the duration of it, how long each waggle run lasts, whether it's a half a second or several seconds, indicates the distance. So it's a lot like the bee making a miniature flight out to the food source.

Let's look at this waggle dance. Okay. You can see that as this bee walks forward she's wagging her body back and forth. In this case she's pointing. Well I should explain that this is a bee doing the dance on the side of a swarm. So it's a vertical surface. And the bee is dancing. As she walks forward doing the waggle motion, she's pointed 90 degrees to



left of straight up. And again, this is on a vertical surface, the side of a swarm of bees. So she's telling other bees the direction and as we'll see in a second, 90 degrees to the left of straight up means the direction of the flowers is 90 degrees to the left of the direction of the sun. We'll come back to that in another diagram. The other thing I want you to notice now is that she's consistent in the direction she faces, as she does the wagging, and she's consistent in how long each waggle run lasts. In this example, each waggle run lasts about eight tenths of a second. And so that indicates a food source about 800 yards away.

Okay, let's take a look at how this information is encoded. As I mentioned, the direction is indicated by the angle of the waggle run. And here we've got an example where the bee is, as she does her waggle runs, you can see that she would be walking upward on the comb but 40 degrees to the right of straight up. And what she's telling the other bees following her dance is that when they go outside of the hive, the flowers are 40 degrees to the right of the direction of the sun. So inside the hive, up is the reference direction. Outside the hive, the sun's direction is the reference direction. So it's a very simple translation from what the bee experiences in the hive to what direction she should fly outside the hive. And then the distance is indicated by the duration of the waggle run. You can see that in the chart. So our little bee that was dancing eight-tenths of a second you can see that corresponds to a distance of about, well I see, 500 meters really. And it's just like doing a long flight. If the food source is a long ways away, each waggle run lasts for longer and longer amounts of time.

So, that's how the waggle dance works. And you might sometimes hear about two types of dances being done in the hive, a round dance and a waggle dance. That's what people used to think but we now understand that these round dances are just waggle dances where the waggle portion of the dance is very very short. It's the same signal, it just looked different before people had means of video recording the dances and studying them closely.

A waggle dancer uses scents, vibrations, and repetition to enhance her communication. A successful forager performing a waggle dance will carry on her body the scent of the flowers she has found; this helps other bees find the exact flowers to which she is guiding them. The number of waggle circuits communicates the quality of a food source. A dancing bee produces only a dozen or so circuits if the food source that she has found is good but not great, but she produces hundreds of circuits if it is a great food source or if the colony is desperate for what she has found (nectar, pollen, water, or even propolis). Her



excitement about the food source is also communicated by how rapidly she conducts her dance.

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Watch: Honey Bee Dances — The Shaking Signal

Like most animals, bees need sleep. Foragers that aren't busy foraging can be found fast asleep on the margins of the cluster or up in the honey supers. You can tell a bee is sleeping because she is very still on the comb. If you poke at her with your finger, she doesn't react. Researchers have found that the brain activity of sleeping bees is reduced, just as it is for people.

During a honey flow, most foragers are at work bringing in nectar and pollen all day, and they only sleep at night. When nectar sources are scarce, however, or when the weather is bad, foragers have little to do. It is then that you find sleeping foragers while doing a daytime hive inspection. During this time there are also many foragers inside the hive that are awake but aren't participating in any tasks.

When a bee finds nectar-laden flowers early in the morning, after a dearth, or following a period of bad weather, she wants to recruit sister foragers, many of whom may be asleep or are not already occupied with tasks inside the hive. She must therefore motivate some of them to get a waggle dance audience, which she does by grabbing inactive bees with her forelegs and shaking them. This shaking motion is accomplished by the bee vibrating her abdomen. Dr. Tom Seeley calls this behavior the “shake and wake.” The foraging bee will first unload her nectar, shake a few inactive foragers, and then fly back out to collect another load of nectar. She repeats this routine a number of times before she has enough of an audience to perform her waggle dance. After the waggle dances begin, it is only a very short time — about an hour — until many foragers may be found at the food source. Most often, the shaking signal is performed on the waggle dance floor. The signal “activates foraging preparations,” so the earliest foragers motivate others to receive information from dancers.

The shaking signal doesn't only occur during foraging; for instance, workers shake the queen during swarm preparation, inducing her to stop egg laying and to become more active on the comb. It is thought that the shake signal is a general one used to alter what bees are doing. In this way, the shake signal means "Switch tasks!" In the context of foraging, the shaking signal tells sleeping or inactive bees to get ready to “watch” a dancer and then go forage as well.

In this video, a bee performs the shaking signal.



Watch: Honey Bee Dances — The Tremble Dance and “Stop” Signal

As increasing numbers of foragers bring their bounty back to the hive during the start of a honey flow (abundant nectar supply), they deliver their harvest to receiver bees. If a forager cannot find a receiver bee, she must recruit one using the “tremble dance.” Another dance performed by foragers, specifically nectar foragers, the tremble dance is used to recruit additional receiver bees to unload nectar from incoming foragers at the start of a honey flow. If a forager that has found an excellent nectar source has a hard time finding a receiver bee to unload her when she returns to the hive, then she will perform a tremble dance instead of a waggle dance. She trembles her body and walks throughout the hive so that middle-aged bees (ones older than nurse bees but younger than foragers) can pick up on this communication signal and learn of the need for more bees to function as nectar receivers. Once more bees are engaged in nectar receiving, the nectar foragers will get unloaded promptly when they get home, enabling them to more efficiently exploit the richly flowing nectar sources that they have found. The average duration of a tremble dance is around 30 minutes! Over time, there will be a rise in the number of nectar receivers, so the colony gradually adjusts to handle the rich nectar sources that the foragers have discovered.

Witness a forager performing a tremble dance in this video.

A worker bee (located slightly right of center in the clip) performs a tremble dance.

The “stop” signal

Eventually, as a food site is failing or too crowded with bees, returning foragers convey this information to other foragers in the hive. They will press themselves up against others and give a short, high-toned beeping noise. This beeping sound has the effect of stopping waggle dances if the dancer has been “beeped.” In this way, waggle dances, shaking signals, tremble dances, and beeping help to stabilize and coordinate the flow of nectar and its processing in the colony, and form the basis of the ways in which bees “talk” to one another to regulate food gathering and storage.

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Read: How Are Dances Perceived in the Dark?

Dance followers stand still during the dance and “listen” with their antennae stretched out stiffly at an angle of between 90 to 120 degrees to each other. The dancer’s rhythmically moving abdomen displaces each follower’s antennae. Because followers use their own gravity detectors to determine where they are on the comb, they can calculate the position of the dancer and thus the direction to the resource that the dancer is advertising.

Bees choose particular places on the comb to conduct their waggle dances. Most dances occur on areas of empty cells about 10 cm square — quite a small area considering a hive typically has an overall comb area of five square meters! Motivated foragers congregate at the dance floor to learn from dancers where the best forage might be. It has been discovered that the wax on their “dance floor” has unique vibrational properties that help to transmit dance vibrations to the maximum number of “listeners”! One experiment showed that bees dancing over empty cells recruited three to four times as many foragers as those dancing on sealed cells.

Normally, a dancing bee creates strong vibrations using her uncoupled flight muscles, which are synchronized with the left and right movements of her abdomen during the dance. However, when the dancer first “calls” for attention but does not yet have a following of interested foragers, she does not use her flight muscles. During this early phase of dancing, she is signaling to distant recruits only through vibrations she introduces to the comb with her legs. Every waggle generates a pull on the edges of the cell she is standing on, and that pull sets up a vibration through the wax. Bees looking for dancers can feel comb vibrations at some distance away and, in response, come looking for the dancer.

The comb dance floor can be “tuned” by changes in temperature and by the addition of propolis to the rims of the cells. Vibrational properties of the comb are slightly dampened by the beekeeper’s four-sided frame. On combs where bees dance, the bees often improve the vibrational property of the comb by constructing large gaps between the edge of the comb and the sides or bottoms of the frames. The wax on the dance floor is also chemically marked by the bees, as has been shown by experiments in which the dance floor is cut out and put back into a different place in the hive. The bees search for



the relocated piece of comb before starting to dance again! They do not search for the “old dance floor” for more than a day, suggesting that its chemical signature dissipates.

Several versions of the waggle dance are variously referred to as the round dance, sickle dance, or transition dance. All are variations of the same dance, modified to indicate distance from the hive to the food source. It has been shown that the different races of bees (Carniolan, Italian, German, and Caucasian) have different “dialects” or “accents” of waggle dances that vary in the number of waggles per run. There is also some evidence that certain odors are released by dancers’ abdomens during the waggle dance to help motivate followers. When these substances were experimentally released into a hive, more foragers left to collect food.

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Quiz: Honey Bee Communication

You must achieve a score of 100% on this quiz to complete the course. You may take the quiz as many times as you need to achieve that score.

Please complete this quiz in the course.

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Discussion: Module 4 Discussion and Questions

This course provides the opportunity to discuss content with fellow beekeepers in addition to the course instructor. You are welcome to participate in all course discussions but are only *required* to participate in two specific discussions (Module 2 and Module 6.)

This Module 4 discussion is **optional** and your participation in it does not count toward your final grade. We encourage you to read other students' comments and to respond if you are interested.

Please answer the following questions about **Module 4: How Honey Bees Communicate**:

- How can understanding honey bee communication make you a better beekeeper?
- What questions do you have for the instructor related to the information provided in this module?

Instructions:

- Click **Reply** to post a comment or reply to another comment.
- Please respond in a text format or as a video using the filmstrip icon that is available once you click **Reply**. (If posting a video response, we recommend that you do not use your cell phone as most do not use Flash software, which is required to convert the recording.)

Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [Cornell's policy regarding plagiarism](#) (the presentation of someone else's work as your own without source credit).

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Module Wrap-up: **How Honey Bees Communicate**

In this module, you explored honey bees' sophisticated communication systems, from their complex use of pheromones to their intricate dance languages. Learning to recognize these communication behaviors in your own colonies enhances your beekeeping skills and deepens your appreciation for these remarkable social insects. [Back to Table of Contents](#)



Module Introduction: **Honey Bee Biology**



Have you ever wondered how bees function as living organisms and navigate the world around them?

In this module, you will explore the biological systems of honey bees that are needed for them to survive and succeed in their environments.

Photo credit: Lucinda VanVleck

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Read: The Nervous System

The honey bee nervous system is an intricate network of many nerve cells called neurons. A typical neuron is formed of a “body,” where most of the cell is found, and “processes,” thin filamentous projections that can be quite long (in some mammals, they can be several feet long!). These processes are called “dendrites” or

“axons” depending on whether they receive input or convey output, respectively. The brain of the bee is where most of her neurons are located, and their processes, both axons and dendrites, extend along a “nerve cord” (analogous to our spinal cords) through the bee's thorax and abdomen, passing through a number of neuron cell body clusters called “ganglia”.

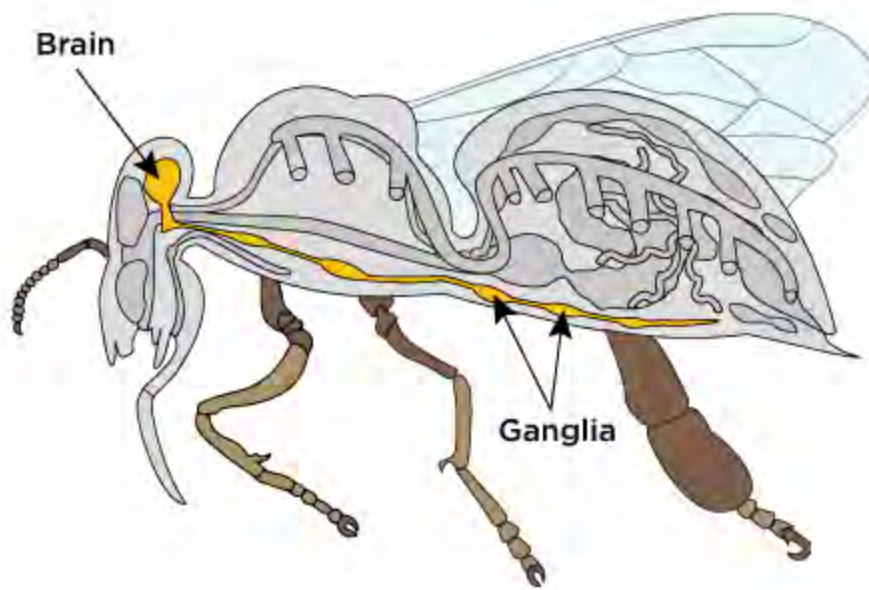
The brain and the ganglia act as “control centers”, connected by nerve fibers to every part of the honey bee's body, including wings, muscles, legs, and even cuticular hairs. Although the bee brain coordinates all the body's functions including the activity of the ganglia, it is the ganglia that exert local control. This is why honey bees can continue to walk, flap their wings, and even sting for a short while after they are beheaded.

☆ Key Points

The central nervous system of the honey consists of the brain and ganglia.

Honey bees have a capacity for learning and memory that surpasses all other insects.





The brain of a worker bee is tiny, and yet it contains nearly one million neurons, so it is not surprising that a worker honey bee's capacity for learning surpasses that of all other insects! They are able to communicate with their nest mates, learn directions, and remember a vast array of smells, sights, and tastes. Honey bees also have impressive mathematical abilities. They are able to count up to four and can understand the concept of zero. After a long foraging trip visiting many flowers, they can determine how to "beeline" home. The bee brain is organized into clusters of neurons that are called lobes. Different lobes of the brain receive inputs from different sensory stimuli, and in turn, each lobe controls a different function or activity in the body. For example, the antennal lobe receives inputs from olfactory stimuli (smells) and the optical lobe receives inputs from visual stimuli.



A very important lobe in the brain is the mushroom body (indicated in orange in the illustration), named so because it resembles the shape of a mushroom. The mushroom body is the center for learning and memory. Sensory information that is processed by different lobes in the brain (like smells and tastes) are interpreted in the mushroom body where it is



learned and remembered. This lobe takes up 20% of the bee brain in volume, while only 2% in a fruit fly. This gives an idea of the vast capabilities of honey bee learning and memory as an insect.

It is useful to realize that the bee's nervous system as a whole operates according to a very simple principle. Information (input) is perceived by sensory nerve cells (neurons) that convey the information to the brain, eliciting a chemical or motor reaction by other neurons (output). This very basic organization “input—(reaction)—output” is found in all animal nervous systems, including that of humans.

The sensory input of the bee's nervous system includes the tactile senses (temperature, pressure, gravity), the olfactory system (smell), the gustatory system (taste), the auditory system (hearing), and the visual system. The neurons concerned with delivering output are either “motor” or “endocrine” and they govern muscle contractions and the secretion of chemicals, like hormones. On the following pages we will take a brief look at some of the inputs and outputs of the honey bee's nervous system.

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Read: Sensory Systems: Mechanoreception

“Mechanoreception” is the ability of an animal to detect and respond to certain kinds of stimuli, such as touch, pressure, temperature, or vibration. Mechanoreception permits animals to navigate their world, react to mechanical stimuli, and even understand their place in space by perceiving gravity. Animals have evolved a clever way to discriminate among different sensory inputs by developing biological structures that are specifically attuned to certain physical stimuli. The most complex of these structures is the eye, whereas the simplest sensory input structure is merely a naked neuron that reacts to being touched or bent.

For bees, sensory information is perceived by many small sensory organs or “mechanoreceptors” that are located on the periphery of their body, just below the cuticle, and on the tarsi (feet), mouthparts, wings, and antennae. These sensory organs are called sensilla (singular “sensillum”). They are small, hair-like structures that contain at least one neuron. Sensilla that perceive tactile information enable honey bees to recognize and learn the feeling of different objects, navigate different surfaces, land on flowers, and physically interact with their nestmates.

▼ Bees can perceive gravity

Bees have groups of sensory hairs called “hairplates” that are essentially a hair attached to a neuron. Hairplates are found on antennal joints, in the neck joint between head and thorax, the waist joint between the thorax and the abdomen, and in joints of their legs. Deflection of these hairs is how sensory neurons attached to them detect gravity. Humans detect gravity in the same way. Hairs located in our inner ears bend when tiny crystals lying over them are pulled by gravity, triggering sensory neurons attached to the hairs.

Bees use their sense of gravity to orient their waggle dance inside the nest so they can communicate resources to other bees. Bees also use gravity to build their combs straight down. This has implications for how level a hive body should be: if a hive is slanted to one side, bees may build comb across the frames, which can become a nuisance for you as a beekeeper trying to move frames. It is important that a beekeeper ensures hives are not slanted



downward to the left or right in the bee yard. Tipping your hive forward to enhance drainage out the entrance is fine, because the angle is parallel to the direction of the combs.



The perception of gravity helps bees build their combs straight.



▼ Bees can hear

When you hear a sound, your ear is detecting vibrations in the air. These vibrations are actually minute changes in pressure, like ripples on a pond, and just like the ripples they occur with different frequencies and amplitudes. Sometimes you may notice that you feel the floor vibrate when someone turns up the bass on their subwoofers. The power of the air's vibrations can be transferred to other substrates. This fact is a key to understanding how hearing works in both bees and humans.

Bees can hear both substrate vibrations (i.e. comb vibrations) and airborne vibrations (e.g. the sounds of dancing bees). They detect surface vibrations with pressure detectors in their legs (more specifically their tibia) called “subgenual organs”, and they discern airborne sounds using another type of sensory hair structure found on their antennae called a “Johnston’s organ”. The quality of the bee’s hearing differs from our human sense, but the basic neural mechanism is very similar.

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Watch: Sensory Systems: Olfaction (Smell)

The sense of smell is arguably the most important sense for honey bees. They rely on smell to help them find and learn important food sources, to communicate through pheromones to their other nest mates, and even to discriminate against non-nest mates that might be trying to enter their nest. Each honey bee colony develops a unique odor profile that is created by individuals secreting hydrocarbon molecules from their exoskeletons. By smelling these hydrocarbons, bees can determine who is a nest mate and who isn't. Watch this video clip to learn more about olfaction in honey bees.

Video Transcript

The sense of smell is called olfaction and honey bees have an incredible sense of smell. They have hundreds of olfactory receptors on their antennae that all bind to different odor molecules. An antenna can even determine the direction of an odor. So whether it's the right or left antenna that's sensing the odor more strongly, that tells them exactly which direction to fly to zone in on a flower. Olfactory receptors are found beneath these hairlike structures called sensilla located on the antennae. And odors bind to these receptors and stimulate a nerve signal to the brain, which can interpret exactly what they've smelled.

When we sequenced the honey bee genome, we found that they have an expansive olfactory system with more olfactory receptor proteins than the other insects than we've sequenced so far. So the fruit fly and the mosquito. So for instance, the fruit fly has 62 olfactory receptor proteins, the mosquito has 79, and the honey bee has an impressive 170 olfactory receptor proteins. And this isn't surprising, given the importance of smell to honey bees. They need to be able to distinguish between their nest mates, and non-nest mates, to smell and interpret all of the different pheromones that are found in the colony, and even to recognize many different types of flowers that they are encountering in their everyday lives.

And honey bees actually have a very good memory when it comes to smells. When the brain is processing the olfactory information from the antennae, they can send the nerve signal to other lobes that help them learn and remember this smell. And one of the major lobes that does this is the mushroom body. It's the center for learning and remembering in the bee's brain. So this is important because they want to be able to remember which flowers are really good nectar sources. So, for instance, maybe in the fall they can smell



that distinct smell of goldenrod and know that this is associated with a really good food reward.

And actually, humans have sort of been taking advantage of this, in training honey bees using classical conditioning, similar to the Pavlov dog. And, what they do is that they expose honey bees to a certain smell, even if it is a smell they would never come in contact with in nature. And then after they expose them to the smell, they'll give them a reward, usually sugar syrup. And honey bees learn really quickly, often after just one or two presentations of this odor food reward system, to associate the odor with this reward. And researchers will actually use this technique to measure learning and memory in bees. And some people have even trained bees to sniff out landmines and illegal drugs using this technique. Though neither of these applications has really proven to be practical yet.

Okay. So researchers can even look at this technique to look at how certain substances might impair learning and memory in bees. So for example, if honey bees are exposed to or consuming a particular pesticide, does this have an impact on their learning and memory? By doing this training, we can see whether or not bees are unable to learn an odor associated with a food reward, or if they learn it more slowly than normal, or if they forget it quickly than they normally would. And even though these sort of presentations for testing, learning, and memory and response to different factors are done in a lab, it actually has practical real world applications because learning and memory is so important for honey bees to navigate in their world, to collect food, and to cooperate socially. So it's just cool to know that by understanding the olfaction system in the honey bee, we're actually able to apply this technique to have broader implications in the bee's ability to function normally in the real world.

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Read: Sensory Systems: Gustation (Taste)

The sense of taste is related to olfaction. In bees, as in humans, the sense of taste is less well-developed than the sense of smell. Bees have about 10 gustatory receptor sensilla types on their mouthparts, tips of their antennae, and tarsi that are specialized to taste a few chemicals, most of them sugars. Humans have around 10,000 taste buds that detect about six tastes. But the most impressive gustatory system that we know of is found in the catfish, which has an amazing 175,000 or more taste buds scattered all over its body!

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Watch: Sensory Systems: Vision

Honey bee vision is well adapted to enable these insects to perform all the functions necessary to thrive. They are able to detect the movement of predators, fly safely, find flowers and other resources, navigate based on polarized light, and use visual cues to find their way back to the hive. Drones also use their eyes to find queens on their mating flights. However, inside the dark hive, bees rely more on their senses of smell, touch, and vibration detection than on their visual sense.

Video Transcript

Honey bee vision takes place in two types of eyes. The first type are the ocelli. These are three eyes that are formed in a triangular pattern near the top of the bee's head and here you can see the middle ocelli. The ocelli detect light intensity, they can't form an image. But detecting light intensity is needed for orientation during foraging, and also to regulate behavioral and hormonal changes that occur between daytime and nighttime. Each ocellus has a lens with retinal cells beneath it.

The next type of eyes are the ones that we commonly refer to as bee eyes on the side of the head. And these are compound eyes and they're much more complex. They're actually comprised of thousands of individual, independent facets with lenses. And this doesn't mean that honey bees see thousands of repeated pictures. But the lenses each form a larger picture, kind of like pixels in a computer screen. The structure of these eyes means that honey bees actually have a low visual resolution. They can't really see great at distances the way that humans can. But that's okay, because they get really close to the things they need to see, like flowers, the nest entrance, intruders. So the effective resolution is quite high. But anything beyond about two yards is actually blurry for a honey bee. Drones have a bit better vision. They have twice the number of facets and lenses as worker bees. Why do you think that's the case? It's to find the queen in the drone congregation areas.

So vision is really important for honey bees because it allows them to detect color, to detect light polarization and also to detect movement. Perception of color occurs with the compound eyes. This is what's detecting the color of light that's entering it. And the colors that bees see are a little bit different from the colors that humans see. Human eyes can detect combinations of red, green and blue. They can essentially see these colors on this



wavelength spectrum. But because bees have different pigments, they can see combinations of green, blue and ultraviolet. Ultraviolet being beyond violet or beyond purple on the color spectrum. So they can actually see these colors on the spectrum. And so you can tell that bees actually can't see red, and they interpret that as black. And their sensitivity to ultraviolet light helps them when they're orienting the sun as a compass, even when the sun is sometimes behind a cloud.

Another neat thing is that because they're sensitive to ultraviolet light, they can also see flowers differently than humans can. So some flower petals actually reflect the ultraviolet light and they appear brighter to bees and they kind of have these really cool patterns on the petals that humans actually can't see. So when they're detecting polarized light, what happens is they're able to detect these bands of light polarization that are in the sky. And basically, air molecules actually scatter particles of light so that you can see this band that's at a 90 degree angle to the sun. And as the sun moves throughout the day so does this band. Always at a 90 degree angle. So as long as bees can see this band of polarized light they can tell where the sun is. And it's a special arrangement of the light sensitive molecules in the compound eyes that enables bees to see where this skylight overhead is polarized and where it isn't. And this is really important, obviously, because it helps bees detect the position of the sun even when it's hidden behind a cloud. But bees must still see some patches of blue sky in order to determine the sun's direction. So if it's completely cloudy, you won't find any foragers about, they'll all be inside the hive.

And the sun serves as a really important visual reference for navigating during flight. They need to be able to tell exactly where the flowers are that they're going to, how to get back to their hive, and especially, how to communicate this important flower patch in relation to the sun to their other worker bees through the waggle dance. Another thing that they're able to perceive with vision is movement. And even though honey bee eyes lack detail, these can actually discriminate fast movements better than humans can. They can discriminate them out of 0.03 of a second, incredibly fast. And this is really important for detecting predators. They can sense any mammals that are moving in front of their colonies so they can have a really quick attack. They could also sense when bee keepers are moving sort of agitated or quickly in front of their colonies too.

A cool thing about movement is that this helps them determine how fast they're flying. So, the speed that objects are moving past their eyes or the optic flow helps honey bees sense how fast they're traveling and the distance that they're actually traveling from the hive, which is also important not only so they can find their way back home, but also to



communicate it through the dance language to their sisters — the other worker bees. Vision is really important for learning and memory in bees. On the first journey from a hive, all bees whether they're drones, workers or queens, they perform an orientation flight. And they're trying to learn what their house looks like from the outside and how they can find their way back home. And they leave their hives and they sort of perform very short flights that are kind of hovering around the hive, and they're taking in what their hive looks like and other landmarks around them so that they won't get lost. And they have to be able to remember this.

Honey bees will use the patterns of polarized UV light and the position of the sun to learn the patterns of the sun's movement during the day. And this is a really cool other thing about memory and it's something that you don't often think about. But for instance let's say that you are a worker bee and there's this really good patch of flowers directly east from your hive. So in the morning when you go and forage on those flowers, you know that you're going to fly directly toward the sun which is in the east. But let's say you come back to the hive and you're just hanging around until the afternoon when you're ready to go back out to that patch of flowers. This time the sun is in the west. So instead of flying directly toward the sun they have to know to fly directly away from the sun to get to those flowers. So bees are actually able to compensate for the change in the sun's position relative to the earth and remember how the sun is moving during the day. They're really good at using it as a directional reference even though it's changing which I find really fascinating.

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Watch: How Understanding Honey Bee Vision Can Make You a Better Beekeeper

Studying an organism's biology is a great way to learn how we can maximize its health and success in the real world. By applying a few simple principles we learned about honey bee vision, we can reduce drift and confusion among our colonies in our own bee yards. These tips are applicable in a variety of beekeeping circumstances from colony arrangement in your yards to moving your colonies specific distances. Watch this video to learn some tips on what you can do.

Video Transcript

Another navigational technique that honey bees use is landmarks. This is really important to know as a master beekeeper because it impacts what you're doing with your colonies. So it's important to have landmarks that are sort of large, clear, and distinctive near the hives. This helps bees find out what their hive looks like and how they can get back home. It could be something as simple as putting the hive near a tree or painting your hive bodies different colors. Or at Dyce, we have sort of these unique symbols that are near the hives to help the bees. So the normal foraging radius or the average foraging radius of a honey bee is about two miles. So they're pretty good at learning a lot of the landmarks within that area.

So if you're going to move your colonies you need to keep this in mind. If you're moving your colonies beyond two miles that's not a problem. A bee will leave her hive, she'll look around, not recognize anything and know that she has to relearn new landmarks and the landscape as she flies about because she's in an unfamiliar place. However, if you're moving your hive within two miles this is when confusion can arise. Now the hive position has changed, relative to familiar landmarks. A worker bee will leave the colony, she'll look around and she'll kind of remember some of the things that she's seeing and then she'll fly back to where the hive used to be. But now it's changed, so they're still recognizing their landmarks, but now they're not able to orient the same way that they did before.

So we recommend to reduce confusion, move your hive small increments of a few feet at a time or move them over winter if you can when the bees are inside for many months and individual bees have to learn their surroundings again come spring. Another way where landmarks are good is in reducing drift. So honey bees drift from their home hive to



another hive, especially if these hives are close together because there's not enough landmarks or visual cues to distinguish among the hives. And this isn't necessarily good, especially with disease transmission or transmitting Varroa mites. So you'd like to reduce drift as much as possible.

And you might think, well isn't that what guard bees are for? Are they going to keep out all of the non-nest mates? And they do keep out a lot of the non-nest mates. But it turns out that if a forager is coming into a neighboring hive full of pollen or nectar, they're more willing to be accepted into that colony. So it's almost like the guard is saying okay, you're bringing food to the party, you can come on in. And so it turns out that there's actually a fair bit of drift that's occurring between colonies. And especially if you have hives in a row, the hives at the end of those rows tend to be comprised of a lot of drifting bees.

And when we think about how hives are actually or how nests are actually spread out in nature in tree kind cavities in these wild bee colonies, they're really spread out. And it's nothing like how humans actually keep honey bees really compact together in an apiary, sometimes with 10 or 20 hives really close together, or even on pallets. So just having extra landmarks so that they can distinguish which hive is theirs is going to help them out a lot.

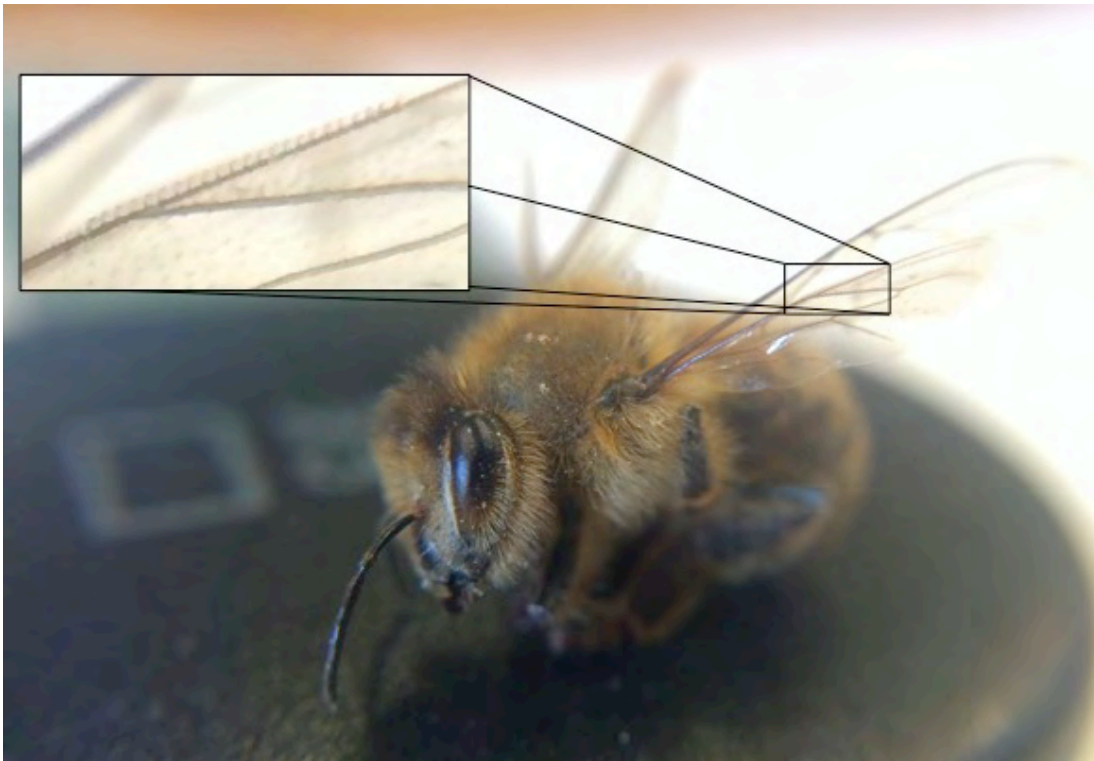
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Read: The Muscular System: Flight

The bee's body has many dozens of muscles that help it move, but by far the largest muscles are those associated with flight, located in the thorax. Honey bees flap their wings incredibly fast (a whopping 240 times per second!) allowing them to reach speeds of 20 miles per hour!

Although they appear to only have two wings while they are flying, they actually have four that hook together in pairs (with hamuli) so they can beat in synchrony and reduce drag and turbulence. Having wings hooked together by hamuli is a characteristic of Hymenopteran insects, as are membranous wings (hence the name of the order Hymenoptera). Veins in the wings add strength to the thin membranes.

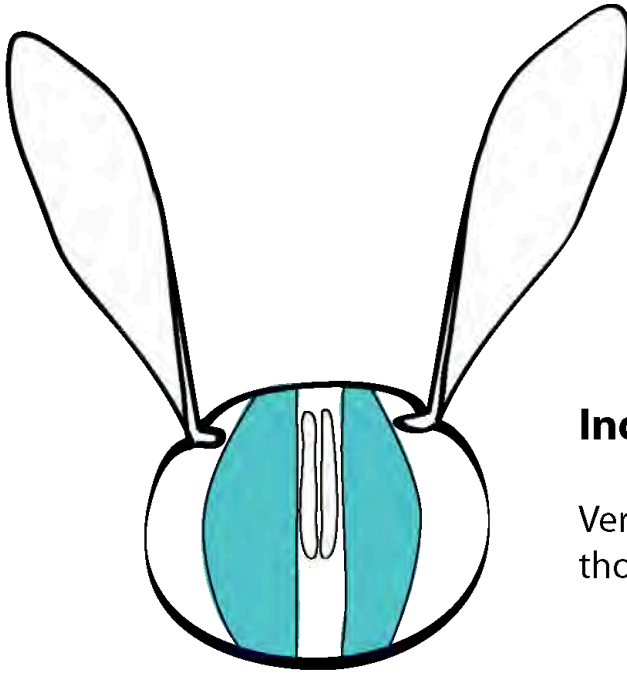


Honey bee hamuli

The way bees and other insects effect flight is one of the most amazing marvels of bioengineering. Bees use both their cuticle and muscles to beat their wings. The cuticle has a tendency to retain its shape, yet it is also somewhat flexible. This allows it to act like a springboard when it is deformed - it has stored energy that is released when it regains its shape. Two massive sets of muscles in the thorax of the bee are attached to the inside of

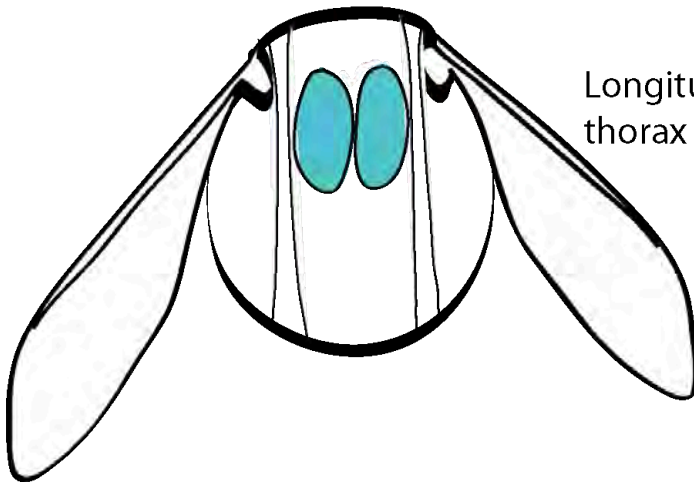


the cuticle as shown in the diagram below. These sets of muscles squeeze the thorax to deform it alternately, either along the body axis, or side-to-side. This causes the wings, which are attached to hooklike wing-processes of the thoracic cuticle, to flap up and down. Thus the power stroke of the wings is aided by the natural recoil of the cuticle.



Indirect flight.

Vertical muscles contract,
thorax flattens, wings rise.



Longitudinal muscles contract,
thorax elongates, wings depress.

Photo credit: Christina Wahl

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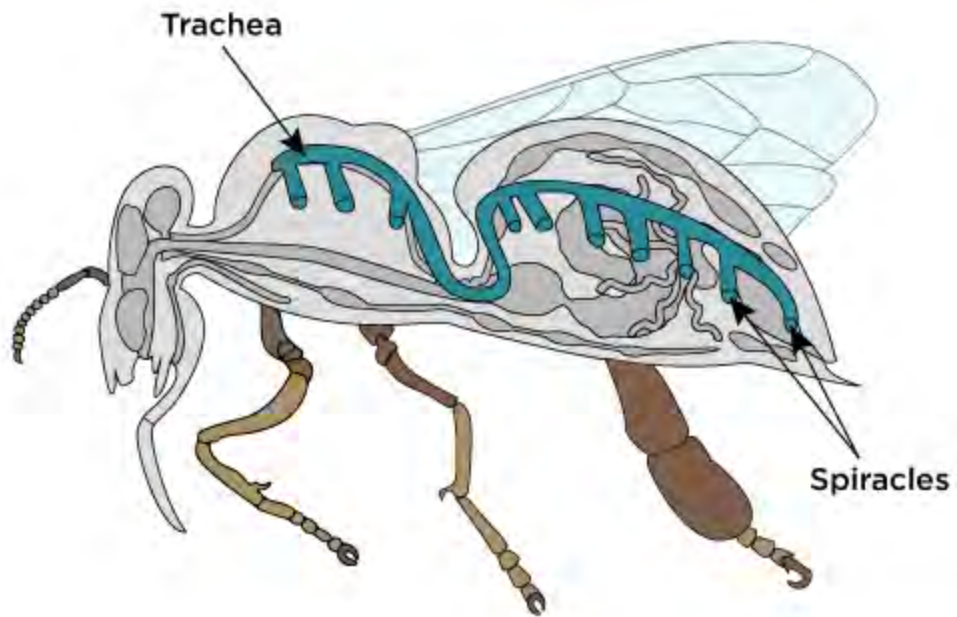
Read: The Respiratory System

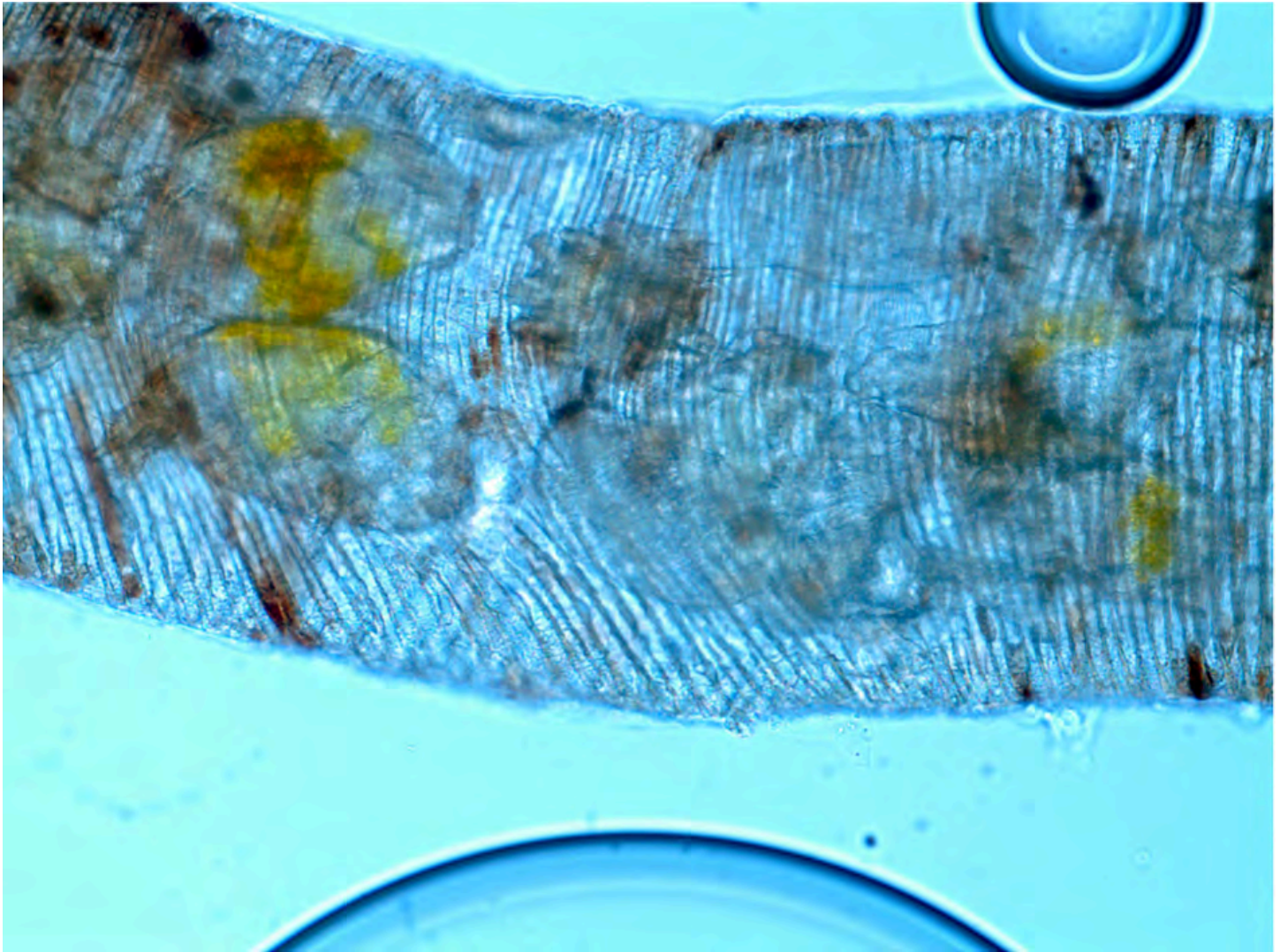
Instead of relying on lungs like humans, honey bees breathe through a series of tubes called tracheae (singular: trachea) throughout their bodies. Tracheae function much like lungs, carrying oxygen to muscles and organs and carrying carbon dioxide away from them.

You can watch a bee breathe. She expands and contracts her abdomen like an accordion. This rhythmical movement brings the air in through the thorax and eventually, out through the abdomen. Air enters the tracheal system mainly through holes in the bee's thorax called "spiracles". From there, the air is drawn throughout the body via smaller and smaller tracheae, much like capillaries. These tiny tracheae come within just a few cells' width of every living cell in the body of the bee, ensuring that adequate gas exchange can occur. After gas exchange occurs, the tracheae consolidate and enlarge again, then exit the body of the bee through several pairs of smaller abdominal spiracles, one pair per segment, for a total of 10 pairs (three thoracic pairs and seven abdominal pairs).

Tracheal mites, a pest of honey bees, enter through the large thoracic spiracle and reside and reproduce in the tracheae, impairing honey bees' ability to breathe. Bees suffering from tracheal mite infections are unable to fly, because the heavy congestion of mites in their tracheae means they cannot get enough air to supply their muscles with the oxygen needed to respire and provide the energy for flight.







Dissected honey bee tracheal tube full of various life stages of the tracheal mite. Photo by Christina Wahl

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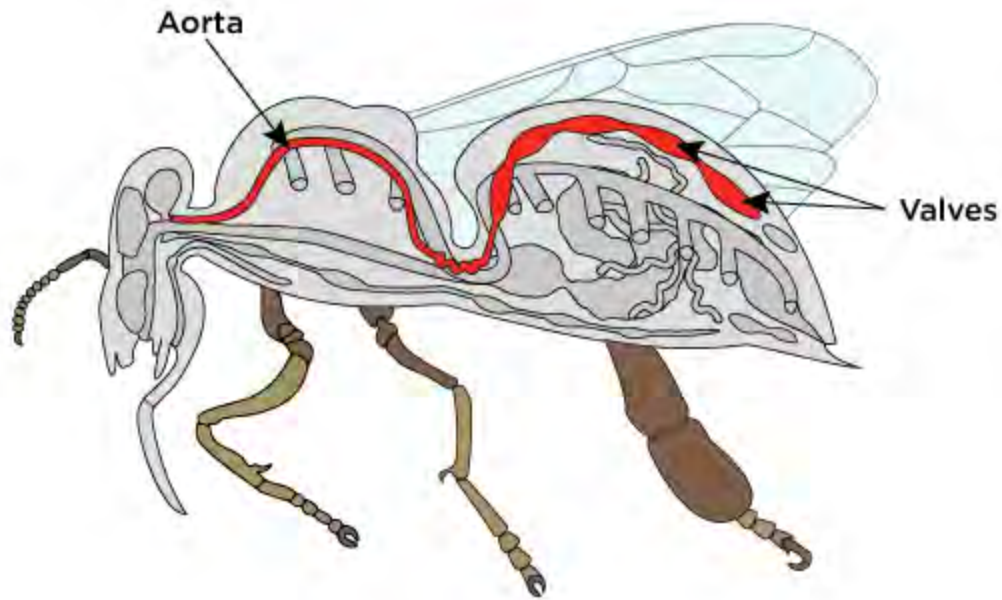
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Read: The Circulatory System

The circulatory system of the honey bee is called “open” because it has no capillaries. In contrast, the circulatory system of humans, containing millions of capillaries, is called “closed”. This is because the blood is contained within the blood vessels of humans whereas in honey bees it leaves the arteries and circulates freely through the bees’ bodies. Humans have fluid moving freely around the body’s cells also, called “lymph”. Thus the blood in the bee is aptly named “hemolymph” since it functions both as blood and as lymph. Hemolymph is kept moving a few ways: 1) by the bee’s heart, located in its abdomen, 2) by the large blood vessels with their muscular walls, and 3) by general body muscle contractions as the bee moves. When bees are inactive for long periods of time – such as during the winter – they can stimulate circulation by contracting the muscles in their abdomens.

Hemolymph that is in the abdomen of the bee enters the heart through the valves, where it gets pumped through the aorta and empties out of the aorta into the body cavity near the brain. As the bee moves, the hemolymph will make its way back to the abdomen. The cycle repeats. In this way, the hemolymph is able to move throughout the bee's body.





The circulatory system has five main functions:

- Transporting energy from digested food to the rest of the body cells to sustain function and life
- Bringing waste from cells to the excretory organs to be removed
- Lubricating body movements
- Conferring immunity through cells that attack pathogens
- Carrying hormones from the glands where they are produced to the organs and tissues they are intended to target

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Read: The Digestive System

The digestive system of animals is like a “tube inside a tube”: Food goes in one end, fecal material comes out the other, and the gut tube’s walls along the way not only function to absorb nutrition and eliminate waste but also to protect the body’s inner tissues from assault by foreign pathogens. Just like an animal's skin or cuticle (the outer “tube” in our “tube inside a tube” analogy), the gut tube functions as a barricade. It is divided into three parts: the *foregut*, *midgut*, and *hindgut*.

▼ Foregut



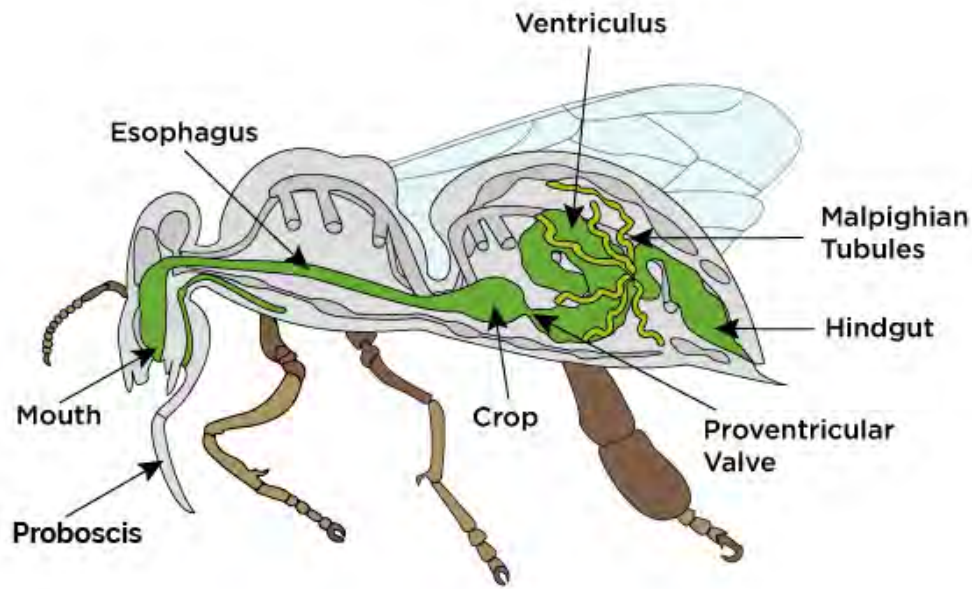
Proboscis

Photo credit: Christina
Wahl

The foregut is located in all three body segments — the head, thorax, and abdomen. Food (nectar, honey, water, and pollen) enters at the mouth, located in the head. Bees use their *proboscis* to bring liquids into their mouth. The proboscis is formed of several mouthparts that the bee brings together to form an airtight tube for drinking. The hairy middle part is what we think of as the bee’s “tongue.” The salivary glands of the bee, which are also part of the foregut, open on its upper surface. Salivary glands are distributed in both the head and thorax. The salivary glands in the thorax are also called thoracic glands. When the proboscis is

not in use, it is folded back under the bee’s “chin.” When she wants to feed, she swings the proboscis forward and brings its several parts together into a tube. The bee then rapidly moves her tongue back and forth, bringing the liquid up through the proboscis and back into her pharynx (throat). If you have a dissection scope at home, you can watch a bee drink. It helps to visualize the process if you give her a colored sugar syrup to drink. During drinking, the outer parts of the proboscis remain stationary, acting simply as the walls of the “straw” through which liquid is drawn in. To chew bee bread, a bee uses its mandibles to break it down into small pieces that it then places in its mouth.





Swallowed nectar, honey, and bee bread pass through the *esophagus* and enter the last part of the foregut called the *crop* (also called the honey stomach), located in the abdomen. The crop is a storage area for nectar, honey, or water. Bees can partially fill their crops with honey before leaving the hive on foraging rounds to use as an energy resource, or they can fill it with nectar or water while foraging to bring back to the hive. A valve called the *proventriculus* is located at the base of the crop; when shut, it prevents the nectar that a bee has collected from moving into the midgut and being digested. The crop can swell enormously from the collection of nectar, displacing other abdominal organs.





A bee that has brought red syrup home in her crop. Photo credit: Christina Wahl

▼ Midgut

While a bee collecting nectar shuts her proventricular valve to keep nectar in her crop, a bee that is eating and digesting honey or bee bread will open her valve so food can pass from the crop to the *ventriculus*, an organ somewhat analogous to both the human stomach and small intestine. Here, food is digested and nutrients are absorbed. The ventricular walls are only one cell thick. They are protected from the abrasive qualities of ingested pollen by an onion-like, noncellular membrane that wraps around and around the inner ventricular walls, called the peritrophic membrane. It is here in the ventriculus that pathogens such as *Nosema* and various viruses break through the peritrophic membranes and infect the ventricular cells.

▼ Hindgut



At the rear end of the ventriculus is another valve, called the ventricular valve, which separates it from the hindgut. The hindgut begins with a short *intestine*, a narrow tube from which the 100 or more *Malpighian tubules* emerge. Malpighian tubules are bee versions of kidneys. They secrete nitrogenous waste from the hemolymph into the hindgut tube for elimination. Little is known of the function of the bee's intestine. It is much shorter and anatomically more simplistic than the small intestines of animals that rely on the small intestine for nutrient absorption.

The hindgut ends with the *rectum*, where solid waste materials are stored for eventual excretion. A wintering bee, unwilling to defecate inside the hive, will store fecal material in the rectum for many weeks. When she is eventually able to relieve herself, the rectum can be enormously engorged, displacing most of the other abdominal organs.



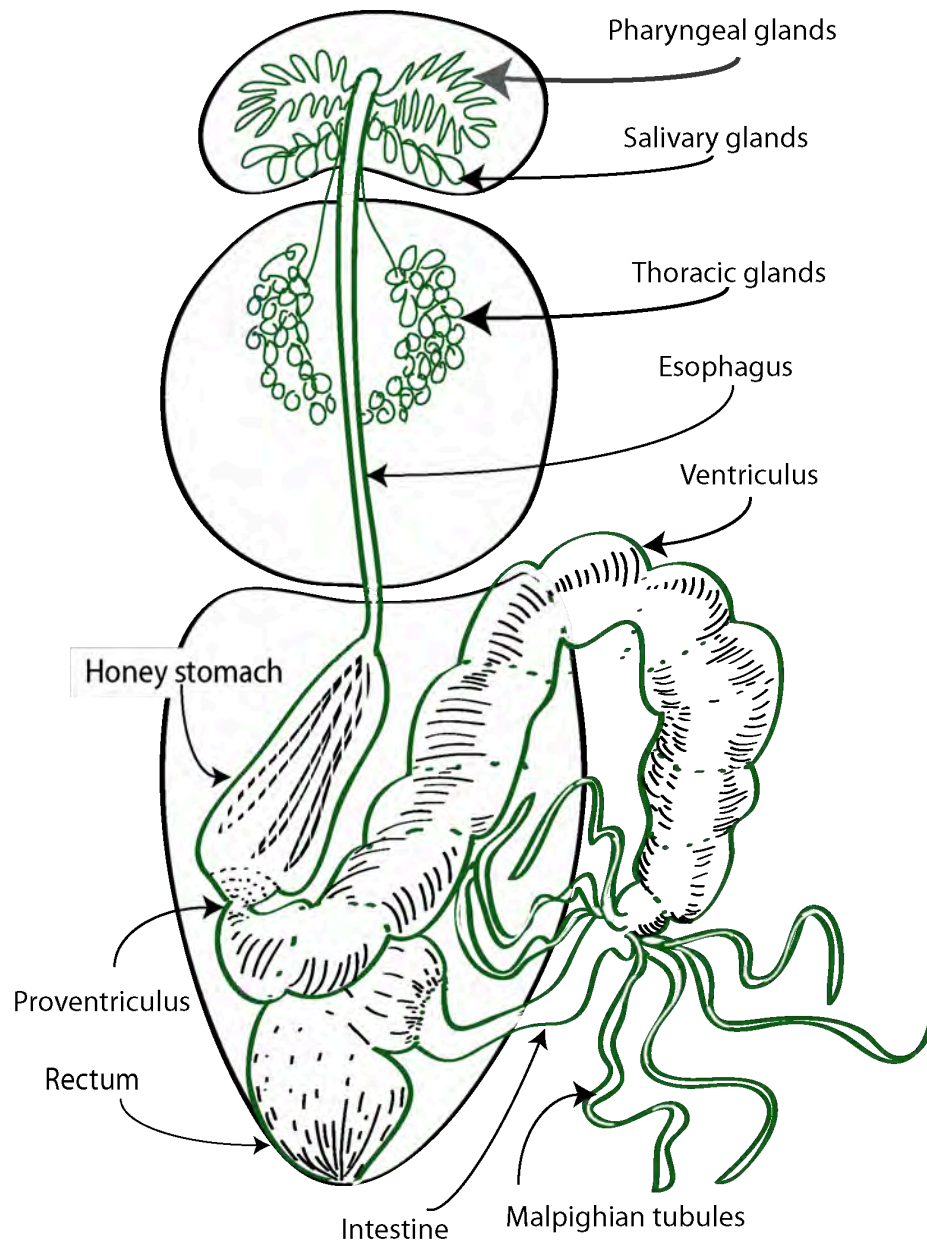


Photo credit: Christina Wahl

Bee feces

Bee feces contain undigested pollen fat and husks; dead cells from the midgut; undigested products from food supplements; some pesticide residues; and, in the case of a disease or parasitic infection, materials that might infect other bees. Bees, therefore, generally do not void waste inside the nest and instead take cleansing flights outside the nest to keep the colony clean and to reduce the potential for disease transmission. The bees' ability to avoid defecation within the nest is especially important in winter, for then they



can be confined inside for weeks or even months. Finding fecal deposits inside the hive or in abnormally high amounts on the hive exterior (dysentery) signals a problem, as dysentery is a symptom of nutritional issues and some diseases.



Orange bee feces on snow from a winter cleansing flight. It is healthy and normal for bee feces to be evident outside the hive in winter, generally after a window of warmer weather when they were able to take cleansing flights en masse.

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Quiz: Honey Bee Biology

You must achieve a score of 100% on this quiz to complete the course. You may take the quiz as many times as you need to achieve that score.

Please complete this quiz in the course.

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Discussion: The Amazing Fat Body

This course provides the opportunity to discuss content with fellow beekeepers in addition to the course instructor. You are welcome to participate in all course discussions but are only *required* to participate in two specific discussions (Module 2 and Module 6.)

This Module 5 discussion is **optional** and your participation in it does not count toward your final grade. We encourage you to read other students' comments and to respond if you are interested.

The Amazing Fat Body

Don't let the name fool you: The honey bee **fat body** is more than just a nutrient-storage tissue; in fact, it plays a major role in colony success. In this discussion, we encourage you to take some time to investigate this incredible organ in beekeeping books and journals, as well as on [Google Scholar](#) and other websites. **In a post, answer one of the questions below and include at least one source to support your response.**

- Describe this tissue and its location in the body.
- What are its functions? How do they compare to the functions of fat bodies in other insects?
- Under what circumstances do bees have large fat bodies?
- Why is the fat body critical to individual honey bee health? To colony health?

Instructions:

- Click **Reply** to post a comment or reply to another comment.
- Please respond in a text format or as a video using the filmstrip icon that is available once you click **Reply**. (If posting a video response, we recommend that you do not use your cell phone as most do not use Flash software, which is required to convert the recording.)

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Discussion: Module 5 Discussion and Questions

This course provides the opportunity to discuss content with fellow beekeepers in addition to the course instructor. You are welcome to participate in all course discussions but are only *required* to participate in two specific discussions (Module 2 and Module 6.)

This Module 5 discussion is **optional** and your participation in it does not count toward your final grade. We encourage you to read other students' comments and to respond if you are interested.

Please answer the following questions about **Module 5: Honey Bee Biology**:

- How can you apply what you've discovered about honey bee biology to improve your beekeeping practices?
- What questions do you have for the instructor related to the information provided in this module?

Instructions:

- Click **Reply** to post a comment or reply to another comment.
- Please respond in a text format or as a video using the filmstrip icon that is available once you click **Reply**. (If posting a video response, we recommend that you do not use your cell phone as most do not use Flash software, which is required to convert the recording.)

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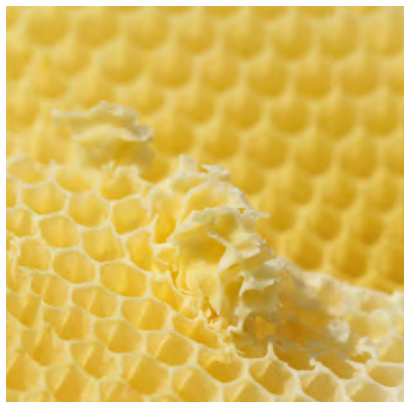


Module Wrap-up: Honey Bee Biology

In this module, you discovered the essential biological systems that enable honey bees to survive and thrive in their environments. This journey through their fundamental life processes showed you how honey bees function as living organisms and interact with the world around them. [Back to Table of Contents](#)



Module Introduction: **How Honey Bees Construct and Maintain Their Nests**



To care for honey bees more effectively, we need to understand their uniquely constructed nests. In particular, the nests of wild honey bee colonies reveal a lot about the bees' needs and preferences, despite regional and racial variations. Much of the information included in this module was derived from studies by Cornell University of wild colonies and swarms being installed in hives.

In this module, you will investigate how bees construct and maintain their homes, what characteristics they prefer, and how they keep them livable and conducive to healthy development.

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Read: The Building Blocks of a Nest

Honey bees are one of the few animals on Earth that are able to make architectural constructions from materials produced by their bodies.

The properties of wax enable it to be an ideal material for building comb, storing honey, and communicating.

Nests are built from wax produced by bees' bodies and plant resins that are processed into propolis.

▼ Wax Production

Wax is composed of over 300 individual compounds and is considered energetically “expensive” for the bees to create. The book "The Hive and the Honey Bee" (2015) estimates that a colony must consume anywhere from five to 20 pounds of honey in order to produce one pound of wax.

Wax production is stimulated by two things: excess nectar and a colony's need for more storage space. Bees *only* build comb under these two conditions. Because comb building is so expensive for the colony, bees typically only produce wax when there is an excess amount of carbohydrates coming into the hive, like during a honey flow or when feeding on 1:1 syrup. Furthermore, the bees will only build comb when they have a need for more storage space; most of the current comb in the hive must be in use before the colony invests in constructing more. Some examples of conditions where bees need more space are during a honey flow, when they need more room to store the incoming nectar, or when the queen is laying plenty of eggs and the colony needs more room to accommodate the growing brood nest.

Worker bees between 12 and 18 days old produce most of a colony's wax (making comb building one of the final roles of house bees before they transition to foraging). Their wax glands (four pairs) are located on the undersides of their abdomens, growing when wax is being produced and shrinking when it is not. If comb building is urgently needed after a bee has transitioned into another job, the bee can reactivate her glands to once again





Worker's wax glands secreting wax. Photo credit: Jennifer Tsuruda

be a wax producer. Liquid wax is excreted from these glands and deposited onto plates called wax mirrors, where it hardens into wax scales or flakes as it dries. Every 12 hours, each bee's gland can form one new scale. The bees then scrape off the hardened wax scales, chew them, add saliva, and manipulate them with their legs. Once the wax is the right consistency, the bees transfer it to the comb, where it is worked over

by other bees to construct comb and cappings.

▼ Comb

Comb-building workers determine how much comb will be built to accommodate worker brood and how much will be built for drone brood. They walk along the brood comb already constructed to evaluate the amount of drone comb that is present. If there is already sufficient drone comb available for the queen to lay in, they will produce worker comb; if not, they will build drone comb. Honey bees will rebuild or repair damaged comb as needed. One behavior that is associated with wax construction and repair is festooning. This is where inactive bees hang between combs, linking themselves together by holding legs with one another. Comb is often repaired beneath the bees that are festooning.



In a hive, wax will darken over time, particularly in the brood nest, where it is littered with pollen, pupal cocoons, larval feces, and other debris. Wax also absorbs lipophilic (fat-loving) substances, which include pesticides, antibiotics, and other toxins. Because this litter and these contaminants accumulate in





beeswax combs, it is a good idea to replace them regularly. Our recommendation is to replace the two oldest frames (or the two with the darkest combs) in each hive body with fresh foundation every year. A good recordkeeping tip used by many beekeepers is to write the year on the top bars of each frame to easily identify which are the oldest to rotate out. You can work these combs toward

the edges of the brood nest so they are not full of brood when you would like to remove them.

Honeycomb is commonly referred to as an architectural marvel: Its shape uses a minimum amount of material for maximum strength and storage space.

The combs themselves are generally parallel and upright in accordance with gravity, with the cells tipped upward slightly to help keep in their contents.



Comb serves not only as a place for food storage and brood production but also as a medium for communication within the hive by transmitting vibrations. The nest interior is dark, so sound and vibrations are important for communication.

▼ Propolis

Propolis is used to caulk and seal holes around the perimeter of the hive, protect the hive from moisture, and even cover/mummify intruders that cannot be removed (e.g., mice). Bees also add a thin layer of propolis to the



surface of newly constructed combs to provide structural support. Notice this layer of orange propolis on the photo of the comb to the right.



Propolis used to seal the hive inner cover to the top hive body. Photo credit: Bee Informed Partnership

In the nests of wild colonies, a layer of propolis often covers all the inner walls of the nest cavity. Propolis even provides the colony with antimicrobial protection, and research has determined that creating a propolis envelope around the nest cavity can actually help protect the colony from some pathogens. This is an important advance in understanding the function of what has long been

considered just a sticky and unsightly annoyance!

Propolis is collected by specialized foragers who carry home resin from a range of trees (poplar, aspen, cottonwood, pine, and others) in their pollen baskets. This resin is mixed with wax to create propolis. It can be sticky or stiff, depending on its origin and temperature.

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Read: How a Nest of a Honey Bee Colony Is Organized — Location and Foraging Area

Honey bee colonies can and do thrive in many locations, including forests, cities, and agricultural areas, in part because honey bees fly so far from their homes to collect food. Estimates vary, but an individual forager may fly up to six miles from the nest, although they usually fly shorter distances when resources are available (one to two miles). Even with foraging trips extending out only two miles, the foraging area of a honey bee colony is more than 12 square miles, or over 8,000 acres. This flight range is important for management: It affects how beekeepers respond to pesticide applications, how many colonies they can place in one apiary, where and how far apart they locate multiple apiaries, whether they must feed closely packed colonies to supplement their forage, and so forth.

Natural areas have long been recognized as prime locations to keep honey bees due to the variety of floral resources. Agricultural areas can also have excellent resources and yield bumper honey crops, but some can put the bees at risk for pesticide exposure. More recently, cities have proven to also be a good habitat for honey bees (and for many other wild bee species), although, of course, this can vary enormously. Cities tend to have high floral diversity from the presence of gardens, parks, green roofs, and community gardens; this and other factors have contributed to the rise in popularity of urban beekeeping. Beekeeping was legalized in New York City in 2010. Beekeepers in urban areas, however, need to be especially cognizant that their bees don't bother neighbors and passersby.

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Watch: How a Nest of a Honey Bee Colony Is Organized — Bee Space



We'll start out small to understand the organization of a honey bee colony's nest. Combs in wild honey bee nests and managed honey bee hives alike are built about $\frac{3}{8}$ " apart from one another, a dimension known as bee space. This leaves just enough room for bees to pass one another on opposite combs. If the open space between two combs is greater than $\frac{3}{8}$ ", then the bees will tend to build comb (commonly called

burr comb) to fill this extra space. If there is a space less than $\frac{3}{8}$ ", bees will likely fill it in with propolis and won't use this space at all.

Bee space has implications for beekeeping equipment. It prevents the combs from getting stuck to each other and to the hive walls, and therefore allows the beekeeper to inspect a colony without needlessly ripping apart the combs. Indeed, the invention of a movable-frame hive by Reverend Lorenzo L. Langstroth depended on understanding bee space. Before designing hives that adhere to the dimensions of bee space, it was nearly impossible to remove the combs to evaluate the health and productivity of the colony without destroying the comb. Being able to pull out each comb one by one without greatly disturbing the colony caused a fundamental change in the way people kept bees. It just goes to show how much beekeepers can learn by studying the biology of honey bees. The patent for the Langstroth hive was filed in 1852, and the hive design spread throughout North America and Europe, ushering in the modern era of beekeeping.

Watch this video to observe how nests and hives of honey bees are typically organized.

Video Transcript

Typically brood is raised in a spherical arrangement in the bottom and center of the nest surrounded by a thin layer of pollen and bee bread. Honey is stored around and above the



brood nest, hence the name of a honey super. Super means over or above. When you pull all frames from a brood nest in a time when there's ample nutrition and ample brood production, you can expect to see ideal frames that have brood in the center with arcs of pollen surrounding the brood, and an arc of honey surrounding the pollen. Seeing frames like this always gives the beekeeper a good feeling that there's lots of resources coming in.

Sometimes however, a colony can become honey bound. This is where a nectar flow is occurring, but there's not enough space for the colony to store it above the brood nest. In these cases, the bees who receive the nectar will deposit it in empty cells in the brood nest, thereby restricting the area where the queen can lay. The cells that are being filled with nectar are usually cells that the queen would lay eggs in. And this can reduce the amount of overall brood that's being produced and if there's not enough available space to continue rearing brood, the colony may swarm. Similarly, colonies can become pollen bound, if there's a very large pollen flow in your region. We'd recommend regularly checking in the amount of pollen or honey in your hives, and where they're being deposited, to make sure that bees have enough space.

If your colony is pollen or honey bound, the beekeeper can make more space in the brood nest by switching out honey filled frames, with frames of drawn comb or foundation. The honey filled frames can then be moved to a higher box if those boxes hold the same size frames or they can be placed in a freezer and then thawed, and put back in the same colonies later in the season when food reserves are low. Reinserting them back into the same colony that they came from will help reduce the transmission of any disease spores that may be present. It's important to note that over time the organization of the hive changes. Bees move up vertically within their nest, so in summer and autumn the brood nest will be located in the bottom deep of the hive with honey stores above. But throughout the winter, the bee cluster moves upward to consume the honey stores. By springtime the clusters occupying the upper deep and there's virtually no stores or bees on the bottom deep.

The colony will become congested if they've got no more room to grow vertically or to store honey. And this has implications for management. Bees tend not to store any honey below the brood nest so beekeepers will often switch the positions of the two deep boxes in spring to bring the brood nest back down to the bottom deep and giving them more storage space above. And this practice of rotating your deeps to ensure the brood nest is brought to the bottom is referred to as rotating hive bodies or reversing hive bodies. And it allows the bees to use their space in a continuous, seasonal cycle. It's really important that



you first check all the frames in the colony to make sure the brood nest isn't split between the two deeps. If that's the case, you can accidentally disrupt the brood nest when you're rotating hive bodies. And this will make it hard for the bees to keep all the brood warm and can result in chilling of the brood.

The colony's vertical movement over time is also a consideration in winter. Honey stores should be located above the cluster going into winter and the beekeeper should remove the queen excluder so the queen isn't left behind as the cluster is moving upwards. The position of queen cups and queen larvae are also created in specific areas of the nest. So within the brood nest queen cups tend to be toward the bottom of the cones or frames and facing downward. Queen cups are present at most times during the year, but are most abundant in the spring swarm season. These special cups are called queen cells once eggs have been laid in them. Queen cells that are produced in preparation for swarming, called swarm cells, are often located at the bottom of the comb.

The exceptions to this general rule are with supersedure cells and emergency queen cells. Supersedure cells are created while the original queen is still alive, but the workers would like to supersede her with a new queen. These cells are the most often found in the center of the comb although there are also exceptions to this as well. Emergency queen cells are built when workers abruptly find themselves without a queen and they must convey larvae that were originally intended to be workers into queens instead. And these cells can be located wherever there are worker larvae. The timing of the production of drone comb is just as important as its location. It's built only after the hive has become well established. The Cornell bee researcher Michael Smith calls drone comb construction a sign that the colony is undergoing puberty. That is, it's starting to invest in reproduction by beginning to produce drones. And these drones are helping the colony reproduce because they're passing on the colony's genes by mating with queens in the region.

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Read: How a Nest of a Honey Bee Colony Is Organized — Nest Characteristics

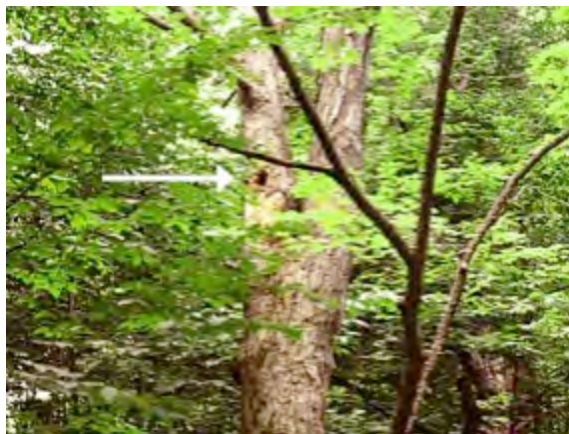
Understanding a honey bee nest characteristics is very helpful for beekeepers. Let's explore the precise architectural elements that make up their homes.

▼ Size/Volume

Wild honey bee colonies living in hollow trees occupy an average nest cavity size of about 10 gallons and range from five gallons to about 15 gallons. This is much smaller than a typical Langstroth hive, in which two deep brood chambers and one medium honey super amount to about 30 gallons.

▼ Entrance

Bees in temperate climates tend to prefer the entrances to their homes to be small (an area only two to three square inches wide), south-facing, high (more than 15 feet off the ground), and positioned at the bottom of the nest cavity. These characteristics help the colony to defend its nest against predators and intruders, to regulate its nest temperature, and to store and ripen enough honey to survive winter in the cold climate of the northeastern U.S. Beekeepers can manipulate hive entrance size with simple tools like an entrance reducer.



Honey bee nest entrance high in a tree. Photo credit: Thomas Seeley

▼ Previous Use

One colony will often continuously occupy a cavity and can do so for decades. Swarms will preferentially occupy a previously used cavity, indicated in part by



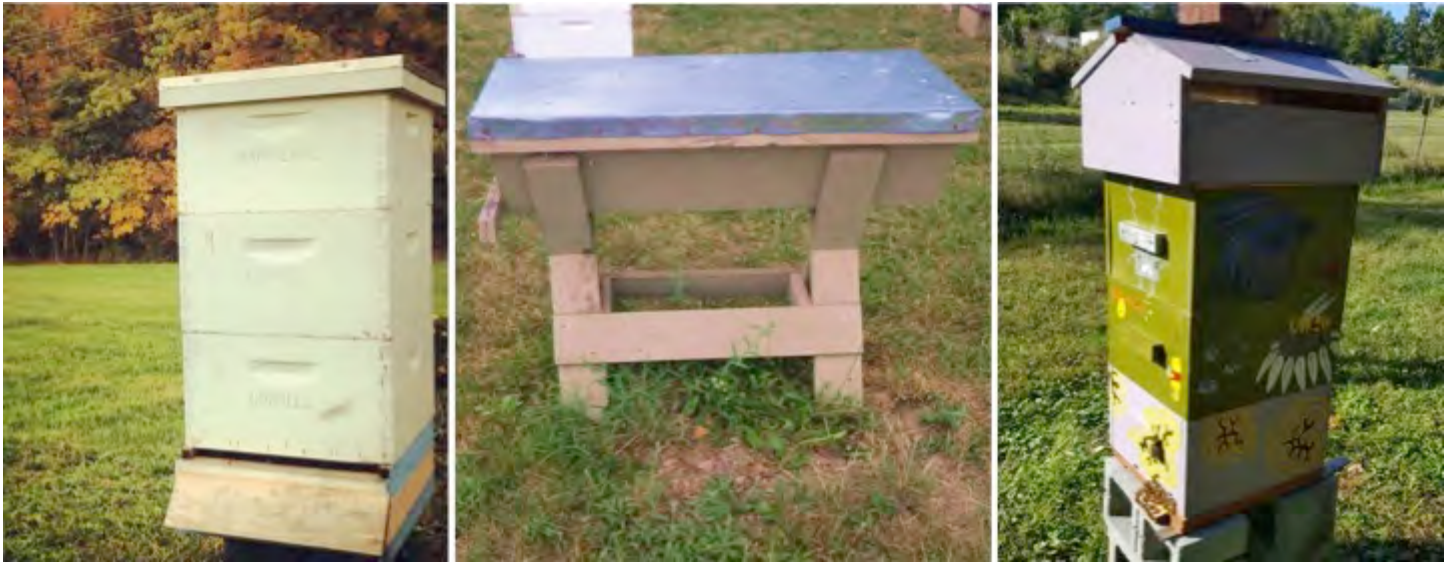
the presence of comb. Bait hives with comb are more successful in luring swarms than those without.

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Read: Hive Styles

The Langstroth hive (photo on left) is the style that we will be discussing throughout the course. There are many variants of the Langstroth hive, which are also popular in Europe, and you can order or build Langstroth hives that fit eight frames or ten. Some other hive styles that you may be familiar with include the top-bar hive (seen in the middle photo and sometimes called the Kenyan top-bar hive) and the Warré hive (in the photo at the right). The best hive style for your bees depends on factors such as climate, goals, and budget, but all hive styles and management regimes require regular inspections.



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Read: Temperature Control in the Nest

Unlike most insects, honey bees are capable of generating heat by exercising their flight muscles, both during flight and when they are stationary inside their nests. Given the ability of individuals to generate heat, it is not surprising that a whole colony of honey bees is able to keep the center of its nest (the brood area) at a constant and elevated temperature, approximately 92°F to 97°F (33°C to 36°C). This ability to regulate the temperature of their environment is called **thermoregulation**. Thermoregulation helps the colony's brood to develop properly; temperatures outside of this range lead to developmental and behavioral abnormalities or failure to emerge as adults at all. This capacity for regulating the temperature of the nest has enabled *Apis mellifera* to thrive in a wide range of climates around the world, including the cold temperate regions of Europe and North America.

Bees can regulate the temperature of their nests in two ways: passive regulation and active regulation. Passive thermoregulation refers to temperature control effected through the structure of the nest itself, while active thermoregulation is temperature control effected through behaviors performed by the bees. Bees expend energy performing active thermoregulation.

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Read: Temperature Control in the Nest — Heating

Honey bee colonies use several techniques to keep the brood nest portions of their nests warm. This allows them to be active inside the nest and ready to fly in cool weather without first requiring the sun to heat the nest from the outside. This also allows them to survive and even rear brood when the outside temperature drops to frigid lows during winter.

Passive strategies to maintain a warm brood nest include nesting in cavities whose walls act as insulation and shelter them from cold winds. The entrance is usually near the bottom of the nest cavity to keep in warm air (as hot air rises), and cracks are sealed with propolis to keep out cold drafts. Wax and honey that surround the brood nest also provide additional insulation.

Active thermoregulation largely pertains to the incubating of brood. Heater bees cover the brood nest, and some may press their warm bodies against the capped brood, increasing the temperature much in the same way that a chicken incubates an egg. Others crawl into empty cells headfirst to heat adjacent brood from the side. In these ways, the brood is kept at approximately 92°F to 97°F. Beekeepers should take care not to open up the brood nest in cold weather, exposing and chilling the developing bees. We generally recommend not to expose brood to temperatures below 59°F, and to not expose brood to rain, which can cause them to become chilled and fail to develop.

The most notable active thermoregulatory behavior that bees use to keep warm is the formation of a **winter cluster**. When ambient temperatures drop below 57°F, the colony clusters together as a group to stay warm. The brood nest is warmed, but the rest of the hive or tree cavity interior is not. While the winter cluster is not technically in hibernation, many other colony functions, such as foraging and water collection, cease when bees are clustering.

Photo: The winter cluster.

Bees on the inside of the cluster generate heat by shivering. They contract and relax their flight muscles without flapping their wings. At the center of the cluster, bees can move around, tend to brood (if there is any), and eat honey. Bees around the edges of the cluster act as insulation to prevent the heat from dissipating. Bees take different roles in the cluster at different times, moving from the center to the edge and back over hours or





days. If there is a respite in the cold, individual bees may leave the hive for brief cleansing flights, often their only opportunity to defecate for several months. The cluster as a whole will also move upward to the honey stores when the temperature is warm enough to give them access to food. If the temperature stays too cold for the bees to break the cluster and move, the bees can starve despite there being plentiful honey stores in the nest.

Finally, an interesting note is that the heating behavior of bees can be lethal to predators. The Asian giant hornet is a predator of honey bees in Asia, and bee stings do not penetrate the hornet's cuticle. Instead, bees will perform balling behavior as defense instead. They crowd around individual hornets and heat them to death!



Honey bees will overheat the Asian Giant Hornet to kill it.

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Watch: Temperature Control in the Nest — Cooling

Honey bees have also adapted several techniques to help cool their nests on hot days. Watch the video to learn some of these strategies.

Video Transcript

On hot days, bees also need to be able to cool the hives so that it doesn't exceed a rough ceiling of 97 degrees. Above that point, the heat can kill the brood and adults and can soften beeswax combs. Most cooling of the colony is active, the main passive mechanism is the ventilation that's achieved through the entrance of the nest and any upper entrances provided by the beekeeper. Bees use a variety of active methods to reduce heat. First, just as clustering helps the bees keep themselves warm, dispersing helps them cool the colony. So on warm days bees will actually move away from the brood to other areas of the hive, to help dissipate the heat.

Bees can also fan their wings to increase ventilation within the hive. They gather along the inner walls of the hive and along the entrance and fan their wings to expel warm air from inside. The motion of the wings is actually different from when they're flying. They are moving air horizontally, as opposed to vertically. Fanning bees face the hive entrance and point the tips of their abdomens downward, and then fan their wings. This is in contrast to when they're fanning the nasonov pheromone to orient other bees to the entrance. In this case, the abdomens are pointed upward. So fanning actually helps create air currents that bring in cool air to circulate through the hive.

If fanning alone doesn't cool the nest adequately, forager bees collect water for evaporative cooling. They pass the water off to water receiver bees who place it on the cappings of brood cells or in open cells. Fanning with their wings then evaporates the water and cools the nest. If the temperature continues rising in the nest, more bees will turn to receiving, which signals to water foragers that they need to increase their water collecting and recruit even more bees to fetch water. Bees can only collect water during the daytime, so during especially hot nights it might be difficult for them to control the temperature inside the nest.

Finally, the worker bees may beard on the outside of the hive, where they literally hang out on the front in a beard shape. Leaving the nest helps increase ventilation and also reduces the amount of heat that's generated by workers inside the hive. Bees may also



beard after miticides are applied or if a fume board is inserted as the bees are making an effort to escape the chemicals. But it's most commonly seen when the colony is overheating.

Honey bees are resourceful organisms that have adapted a variety of behaviors to help them survive in temperatures that exceed the thermal range necessary for brood development and adult survival. Additionally, bees in the wild carefully select nest sites that will help with thermoregulation. These passive and active strategies have allowed them to be successfully managed nearly everywhere in the world.

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Quiz: Nest Construction and Maintenance

You must achieve a score of 100% on this quiz to complete the course. You may take the quiz as many times as you need to achieve that score.

Please complete this quiz in the course.

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Discussion: Module 6 Discussion and Questions

This course provides the opportunity to discuss content with fellow beekeepers in addition to the course instructor. You are welcome to participate in all course discussions and are **required to participate in this discussion to pass the course**. Meaningful participation includes the following options:

- Providing thorough answers to the questions posed by the instructor
- Helping answer other students' questions
- Sharing additional information on the topic based on your prior experiences or knowledge

Please answer the following questions about **Module 6: How Honey Bees Construct and Maintain Their Nests**:

- Studying wild honey bee colonies have enlightened us on the needs and preferences of bees so that we can keep them in hives. What else can we learn from observing wild honey bee colonies that we can apply to our own beekeeping practices? Explore books and online resources to tie in additional research on the topic.
- What questions do you have for the instructor related to the information provided in this module?

Instructions:

- Click **Reply** to post a comment or reply to another comment.
- Please respond in a text format or as a video using the filmstrip icon that is available once you click **Reply**. (If posting a video response, we recommend that you do not use your cell phone as most do not use Flash software, which is required to convert the recording.)

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Assignment: Course Project: Creating an Outline for a Presentation

Scenario:

You have been asked to give a 45-minute presentation based on your beekeeping knowledge and experience. The audience can be one of the following:

- Your beekeeping club
- Beekeeping conference (both beekeepers and scientific researchers are present)

The presentation can be on any subject introduced in this course (i.e., a subject in **biology**, **behavior**, or **evolution**). Choose a topic that you found particularly interesting and use it as a jumping-off point. Conduct additional research using books, reputable websites from scientific organizations or extension personnel, and/or beekeeping magazines as resources to help you find information to include. Also include information about how the subject is applicable to your beekeeping activities.

Assignment:

Create an **outline** for the presentation. You do not need to create a full presentation with slides. Include the following points in your outline:

- Your chosen audience
- Presentation title
- Main points you will make in your hypothetical 45-minute presentation (typically one page of detailed point form notes in length)
- Description of why the topic is valuable and relevant to the audience
- Personal experiences you will share with your audience
- Two questions you anticipate the audience asking along with your responses to both of these questions.
- At least two points from your additional research that applies to your presentation, as well as the references for that information

This project will be graded as "Complete" or "Incomplete." A grade of "Complete" is required for you to pass the course. You can achieve a "Complete" grade by thoughtfully and accurately addressing **all** points outlined in the rubric below.



Your instructor will provide feedback for your submission and is available to answer questions as you work on your outline. If you receive a grade of "Incomplete," you can resubmit your project after incorporating the feedback from the instructor. Only two resubmissions are permitted.

Instructions:

- Complete the course project.
- Save your work as one of these file types: .doc, .docx, .txt, .pdf. No other file types will be accepted for submission.
- To submit your completed project for review and credit, click the **Start Assignment** button on this page, attach your completed course project, then click **Submit Assignment** to send it to your instructor for evaluation and credit.

Your instructor will evaluate the outline based on the rubric below. Students must complete all criteria to receive a "Complete" grade. Through this assignment, the instructor will evaluate your understanding of course material and your ability to communicate topics related to bee biology, behavior, and/or evolution.

Rubric:

The student identified the audience.Yes/No

The content is appropriate for the audience level.Yes/No

The student identified a title for the talk.Yes/No

The topic of the talk is focused on a subject that was introduced in this course (biology, behavior, or evolution).

Yes/No

The main points are appropriate to the topic of the talk.Yes/No

The main points include an appropriate amount of detail for a 45-minute presentation.

Yes/No

The student convinced the audience of the relevance of their talk.Yes/No



The student included a personal experience.Yes/No

The student provided and answered two questions.Yes/No

The questions posed were realistic to be asked by this audience.Yes/No

The student included at least two points of additional research from a magazine, book, scientific journal, or website.

Yes/No

References for additional research were included.Yes/No

Before you begin:

Please consider that this is a professional forum; courtesy and professional language and tone are expected. Before posting, please review [**Cornell's policy regarding plagiarism**](#) (the presentation of someone else's work as your own without source credit).

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Module Wrap-up: **How Honey Bees Construct and Maintain Their Nests**

In this module, you learned about the intricate ways honey bees build and maintain their homes. Your investigation of their preferred nesting characteristics and colony maintenance behaviors revealed how these insects create environments that support healthy development and colony success. [Back to Table of Contents](#)



Thank You and Farewell

Congratulations on completing "Master Beekeeping Part I: Honey Bee Evolution, Biology, and Behavior."

I hope that you now have a deeper understanding of how honey bees evolved and why they behave the way they do. The material covered in this course serves as a foundation to understand and apply the practical, hands-on aspects of beekeeping. I hope it has met your expectations as you support your colonies' survival and success.

From all of us at Cornell University and eCornell, thank you for participating in this course.

Sincerely,

Emma Walters



Emma Walters
Honey Bee Extension Associate
Dyce Lab of Honey Bee Studies
Cornell University

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Downloadable Resources

You may wish to download and save these files for future reference.

- [The Stages of Development](#)
- [Worker Polyethism](#)
- [Kin Selection](#)
- [Bee Anatomy](#)

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Recommended Books

There are more books written about honey bees than any other animal on the planet, besides humans. With so many to choose from, we've compiled a list to help guide you in your readings.

There are no required readings for this course; all examination questions will be based on the online material only. If you are interested in delving deeper into topics we've discussed, here are some great resources:

The Evolution of the Honey Bee Colony as a Superorganism

- "The Buzz About Bees: Biology of a Superorganism" (2008) by Jürgen Tautz
- "The World History of Beekeeping and Honey Hunting" (1999) by Eva Crane
- "Honey Bee Ecology" (1985) by Thomas Seeley
- "The Social Behavior of the Bees" (1974) by Charles Michener

Wild Bees

- "Bees: An Up-Close Look at Pollinators Around the World" (2015) by Sam Droege and Lawrence Packer
- "A Sting in The Tale: My Adventures with Bumble Bees" (2013) by Dave Goulson
- "The Bees of the World" (2000) by Charles Michener

How Honey Bees Communicate

- "Honey Bee Democracy" (2010) by Thomas Seeley
- "Pheromones of Social Bees" (2000) by J.B. Free
- "The Dancing Bees" (1953) by Karl von Frisch

Honey Bee Biology

- "The Hive and the Honey Bee" (2015) by L.L. Langstroth, et al.
- "The Lives of Bees" (2019) by Thomas Seeley
- "Honey Bee Biology and Beekeeping" (2013) by Dewey Caron and Lawrence Connor
- "Bees and Beekeeping: Science, Practice, and World Resources" (1990) by Eva Crane
- "The Biology of the Honey Bee" (1987) by Mark Winston

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Photo Credits for Video Clips

Watch: The Honey Bee Colony as a Superorganism

Photo of bees on frame: Jennifer Tsuruda

Photo of *Varroa* mite on bee: Jennifer Tsuruda

Watch: Eusociality: How Honey Bees Are Truly Social

Photo of queen with blue mark: Jennifer Tsuruda

Watch: Kin Selection

Photo of eggs: Bee Informed Partnership

Watch: The Importance of Wild Bees

Illustration of species diversity graph created by eCornell, based on an illustration from Bryan Danforth

Photo of metallic green sweat bee on leaf: Bee Informed Partnership

Photo of metallic yellow and green sweat bee on yellow flower: Bee Informed Partnership

Illustration of bee life history strategy pie chart created by eCornell, based on an illustration from Bryan Danforth

Photo of *Colletes inaequalis*: Bryan Danforth

Photo of *Megachile* spp.: Bee Informed Partnership

Photo of *Ceratina* spp.: Bee Informed Partnership

Photo of *Triepeolus lunatus*: Bee Informed Partnership



Photo of bumble bee: Bee Informed Partnership

Watch: Mating

Photo of mating sign: Jennifer Tsuruda

Watch: The Stages of Development

Photo of multiple queen cells: Bee Informed Partnership

Photo of one-day-old larva: Bee Informed Partnership

Watch: How Caste Is Determined

Photo of royal jelly surrounding larva: Bee Informed Partnership

Photo of queen cell: Bee Informed Partnership

Watch: How the Worker's Role Transitions Throughout Its Life

Photo of honey bee flying to Siberian squill: Lucinda VanVleck

Photo of honey bee queen with workers: Lucinda VanVleck

Photo of guard bees: Lucinda VanVleck

Photo of fresh comb: Lucinda VanVleck

Photo of honey bee on clover: Lucinda VanVleck

Watch: Honey Bee Dances

Video clip of waggle dance: Tom Seeley

Illustrations of waggle dance communication created by eCornell, based on illustrations from Tom Seeley



Round dance and waggle dance illustration: Tom Seeley

Watch: Olfaction (Smell)

Photo of honey bee antennae: Jennifer Tsuruda

Video clip of proboscis extension response: Geraldine Wright

Watch: How Understanding Honey Bee Vision Can Make You a Better Beekeeper

Photo of hive landmark 1: Eleanor Andrews

Photo of hive landmark 2: Eleanor Andrews

Watch: How a Nest of a Honey Bee Colony Is Organized

Photo of frame with brood, pollen, and honey: Bee Informed Partnership

Photo of honey bound frame: Eleanor Andrews

Photo of pollen bound frame: Bee Informed Partnership

Watch: Temperature Control in the Nest

Photo of hives in a warm landscape: Bee Informed Partnership

Photo of hives painted in various colors: Jennifer Tsuruda

Photo of bearding on the front of the hive: Jennifer Tsuruda

Illustration Sources Cited

Ovary illustrations and age-related worker activity illustrations are created by eCornell, based on those from Winston M. The biology of the honey bee. Cambridge, Massachusetts: Harvard University Press; 1991.



Honey bee development illustration is created by eCornell, based on The Life Cycle of the Honey Bee Family. Distributed by Dadant & Sons Inc. Hamilton, Illinois.

Complementary sex determination gene illustration is created by eCornell, based on the illustration from Gempe, T. and Beye, M. (2009) Sex determination in honeybees. Nature Education 2(2):1

Honey bee mushroom body illustration is created by cCornell, based on Brandt, R., Rohlfing, T., Rybak, J., Krofczik, S., Maye, A., Westerhoff, M., Hege, H.-C., & Menzel, R. 2005. A three-dimensional average-shape atlas of the honeybee brain and its applications. Journal of Comparative Neurology 492, 1-19.

